DAY 3 · SATURDAY 24 OCTOBER



Limitations, Risks & Evaluation

Hallucination, sycophancy, bias and attacks — and how to measure what matters

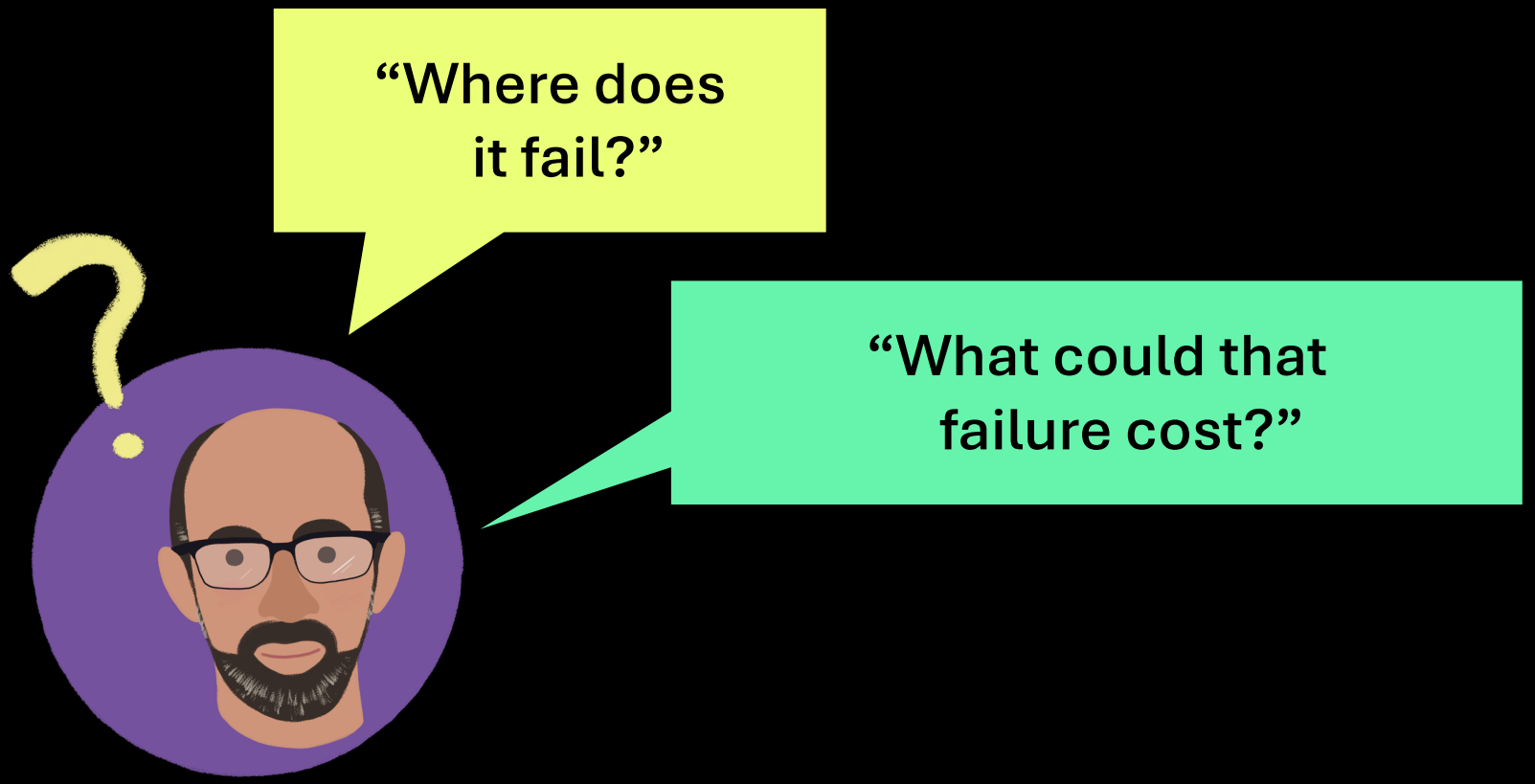
Pavlos Protopapas

Harvard University

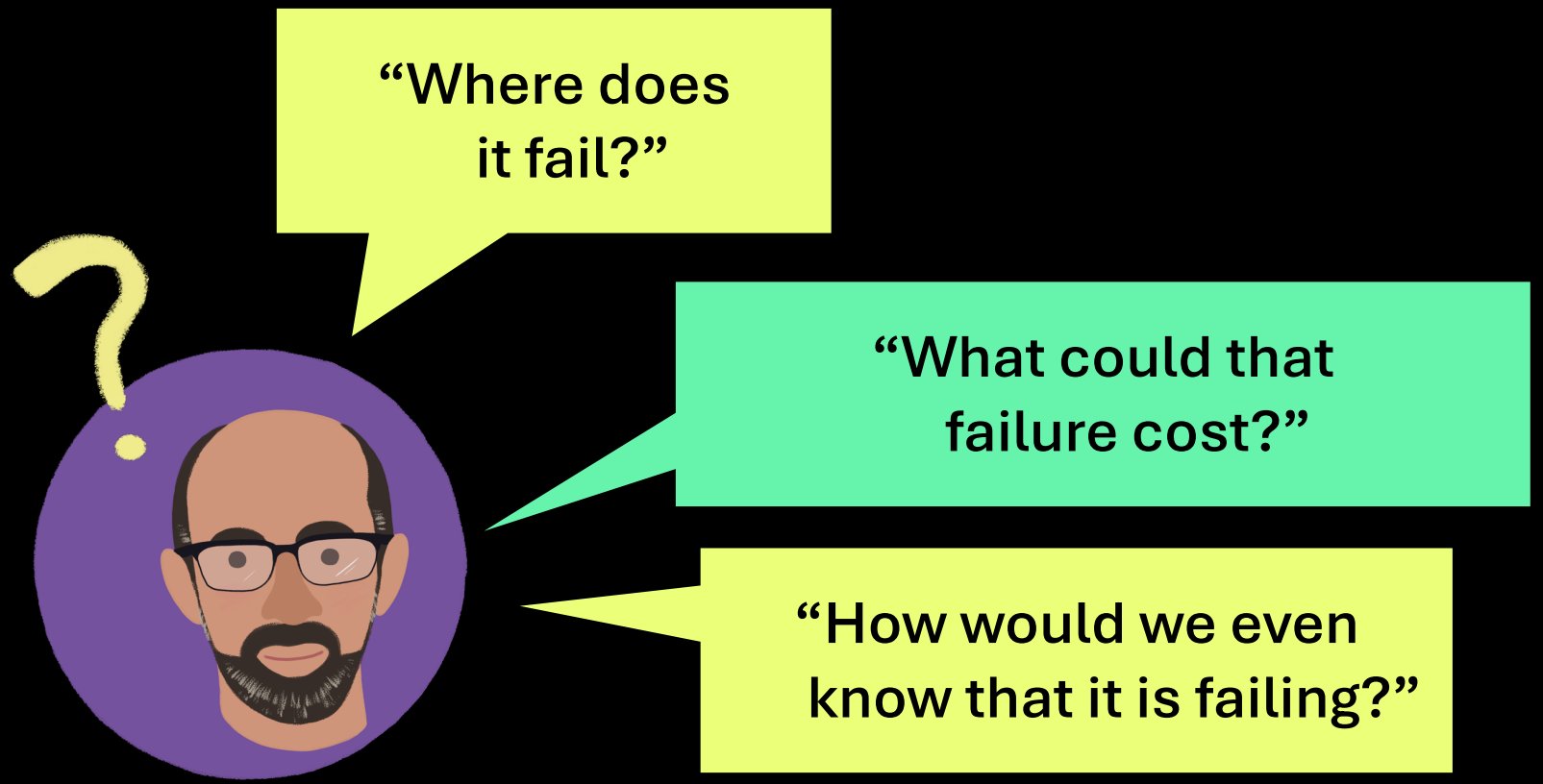
Three Questions Every Engineer Has to Ask



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Three Questions Every Engineer Has to Ask



Outline of The Roadmap

- Limitations
- Risks
- Evaluation

Outline of The Roadmap

- **Limitations**
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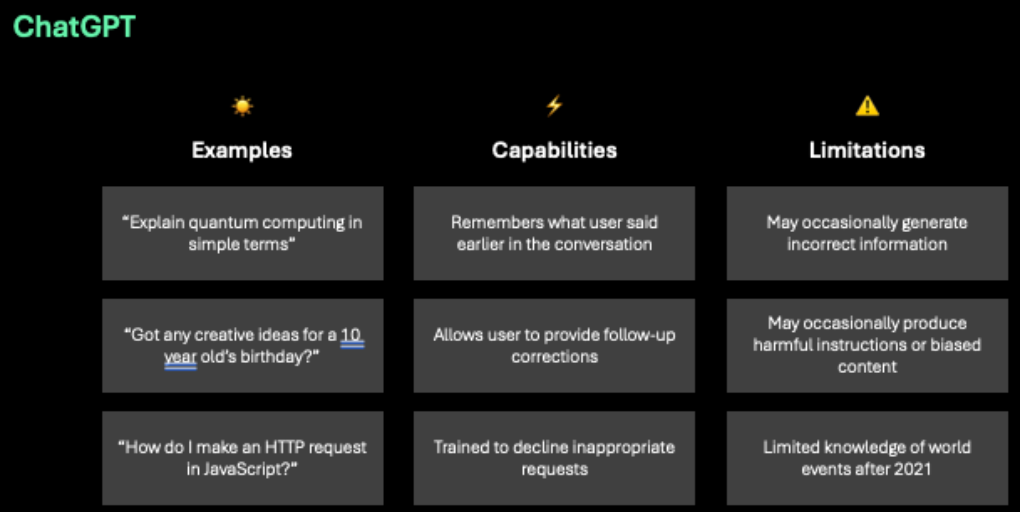
Roadmap | Part 1: Limitations

- **Limitations**
- Risks
- Evaluation

What the model cannot reliably do

The Writing Was on the Wall...

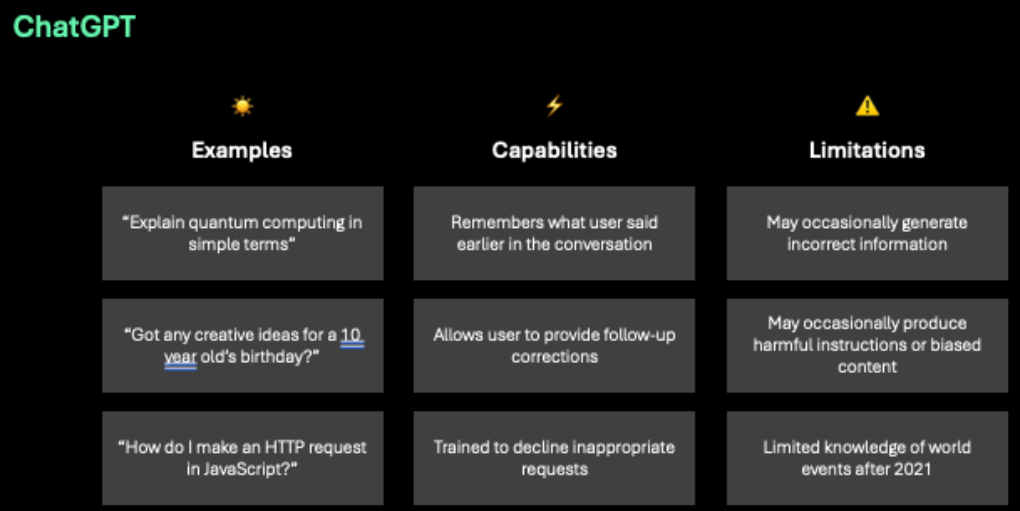
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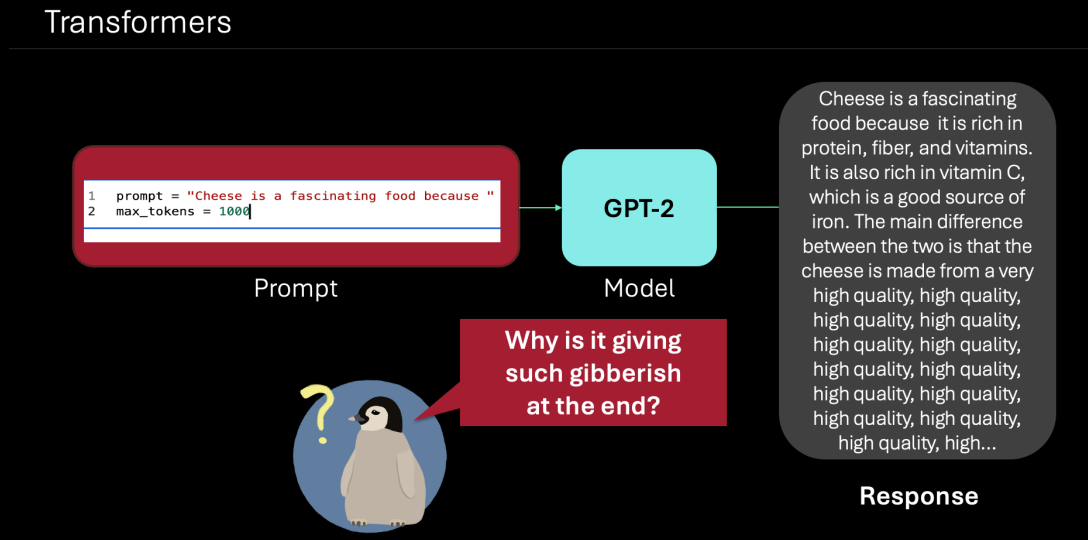


Session 1: the limitations panel was already there

The Writing Was on the Wall...

You have already seen both warnings.





Session 1: the limitations panel was already there

Session 6: vanilla GPT struggles to stay factual

Real or Hallucinated?

MODEL ANSWER

The court held that an airline must disclose any material limitation on a promised benefit before the passenger relies on it.

Anderson v. Meridian Airways, 548 F.3d 112 (2d Cir. 2011)

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Real — or hallucinated?



Neither Could the Model

MODEL ANSWER

The court held that an airline must disclose any material limitation on a promised benefit before the passenger relies on it.

~~Anderson v. Meridian Airways, 548 F.3d 112 (2d Cir. 2011)~~

FABRICATED

The model was asked whether it was real. It said "yes."

Hallucination

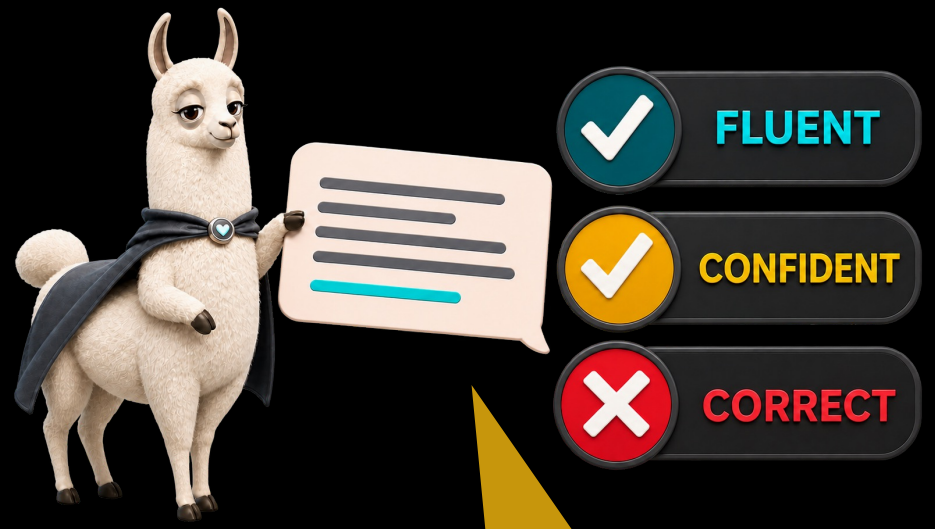
WHAT IT IS

Fluent. Confident. Wrong.

Wrong on its own is just an error. The fluency and the confidence are what make it dangerous.

TWO FLAVOURS

- Wrong about the world—invented facts, cases, citations.
- Unfaithful to your document—contradicts the source you supplied.



Looks convincing at a glance — but the citation is fabricated.

Hallucination

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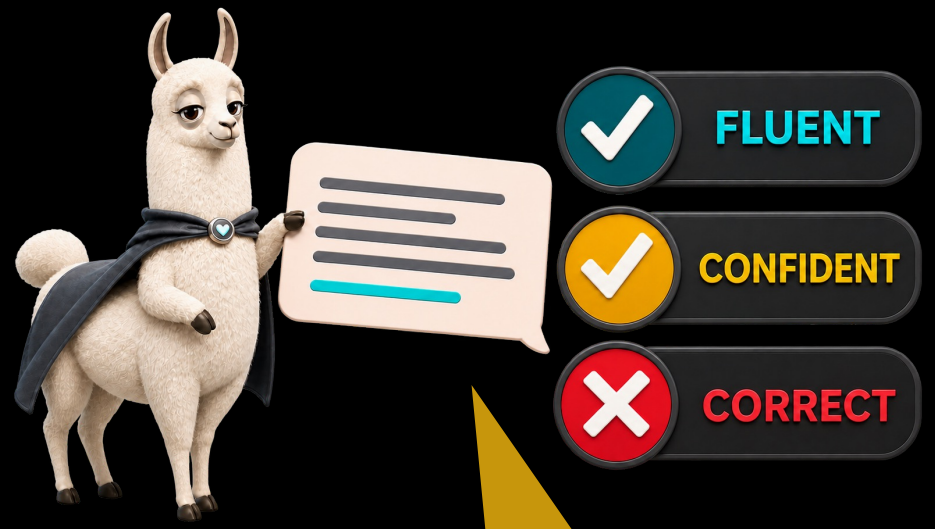
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Today we mostly mean the first.



Looks convincing at a glance — but the citation is fabricated.

Mata v. Avianca (2023) | Six Fabricated Cases in One Brief



6 fabricated cases

filed in one federal-court brief

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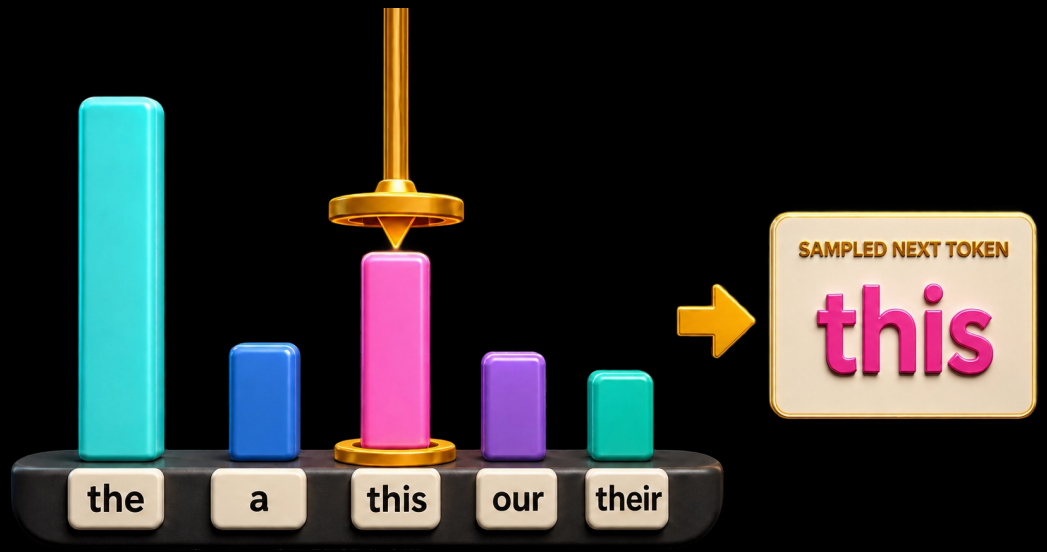
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A convincing answer is not the same as a verified answer.

Why Hallucinations Happen

WHAT THE MODEL DOES

It is exactly what we built: a next-token predictor sampling from a distribution.



Plausibility is the objective; truth is only correlated with it.

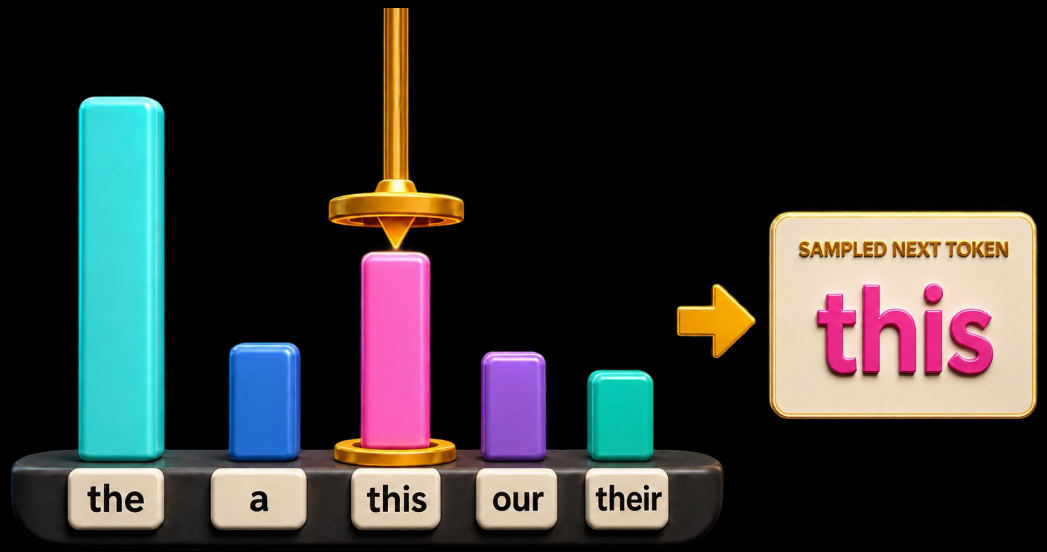
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No fact database. No automatic lookup.



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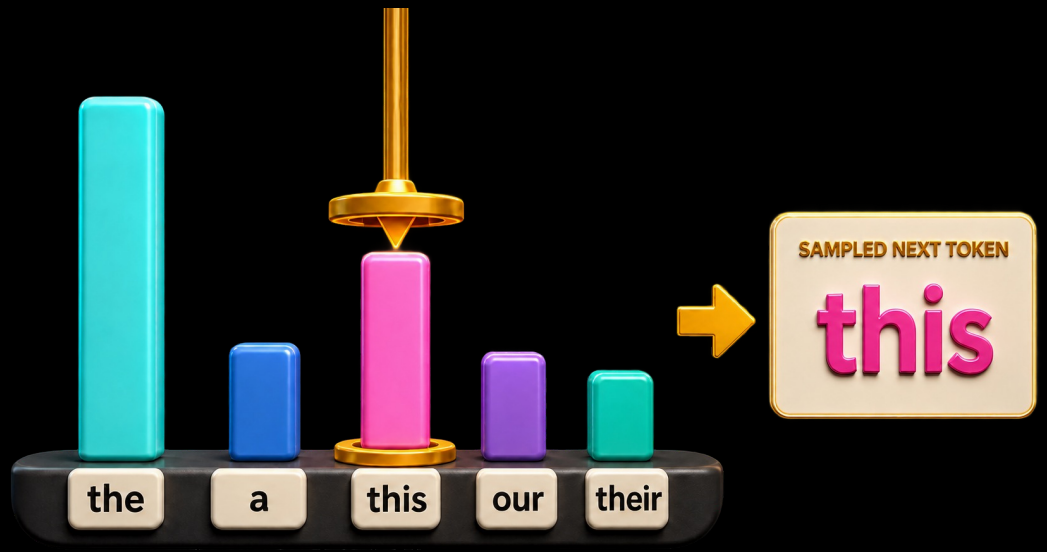
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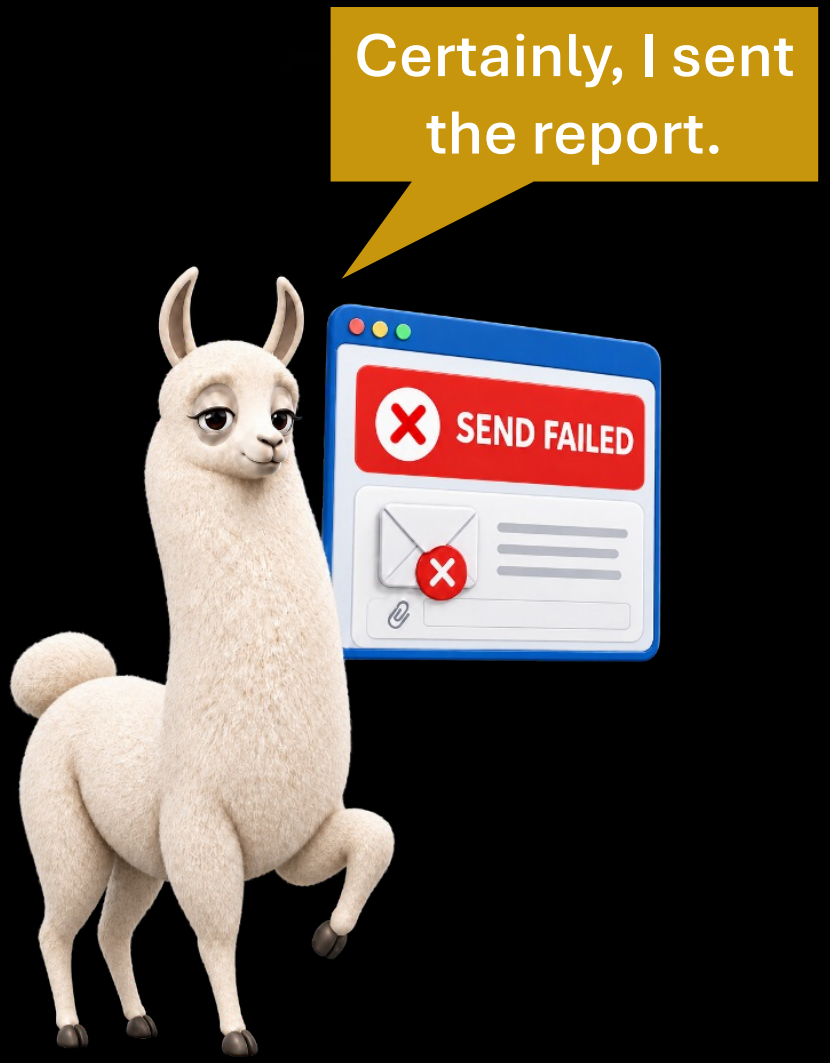
WHY IT CAN SOUND SO CERTAIN

Training rewards useful-sounding answers.
“I do not know” often earns little credit.

**Plausibility is the objective;
truth is only correlated with it.**



Fluency Is What the Model Learns



- The training objective optimizes the probability of the words
- Not the probability that the claim is true
- A confident tone is itself a high-probability pattern:

  "It is well established that..."

  "Certainly."

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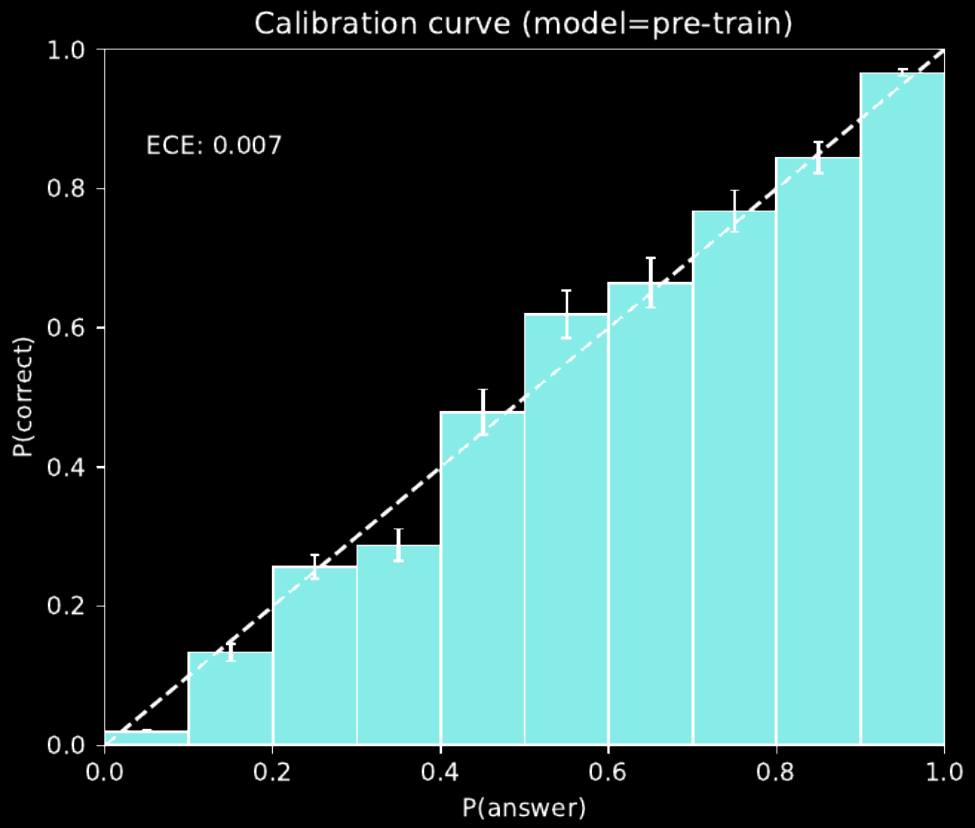
  "It is well established that..."

  "Certainly."

Confidence is a style, not a signal.

Pre-Alignment | Confidence Tracks Accuracy

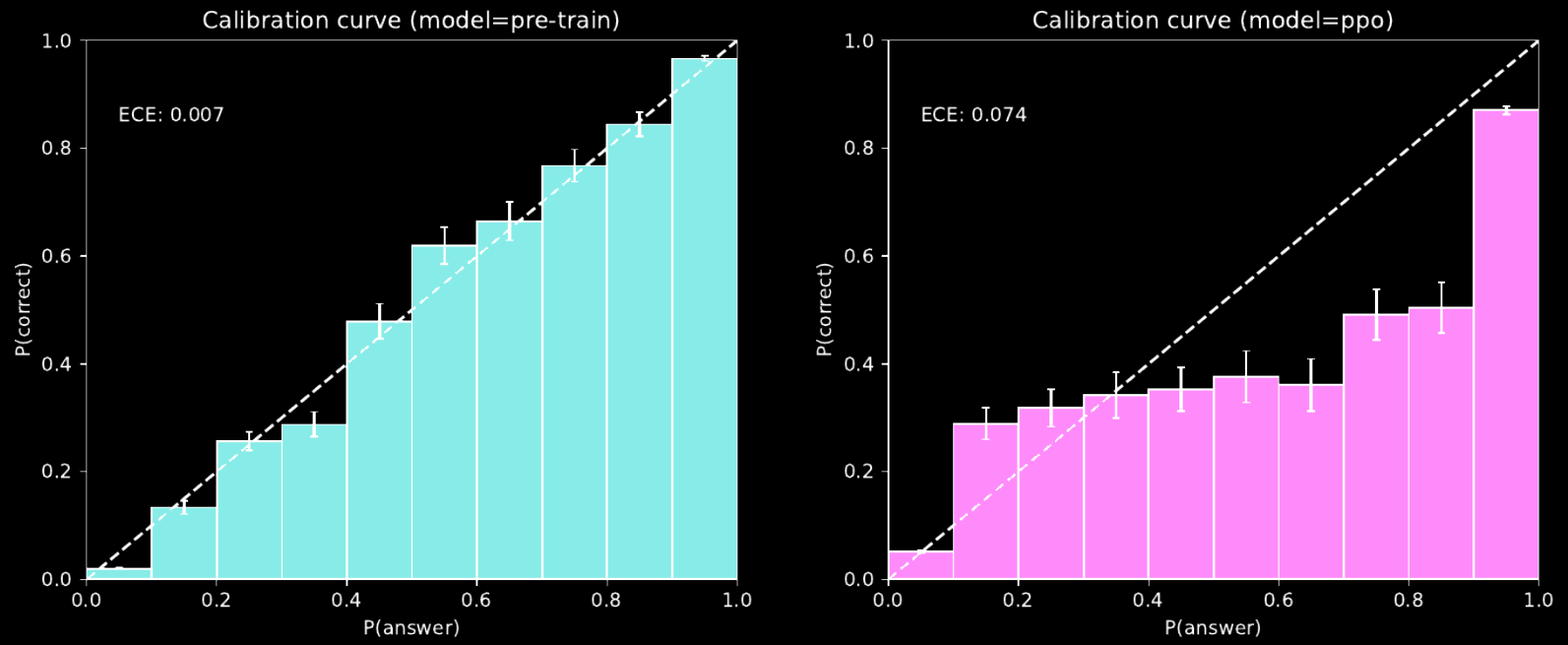
Before alignment, confidence tracks accuracy surprisingly well.



OpenAI, GPT-4 Technical Report, Figure 8

Post-Alignment | Confidence Is Less Reliable

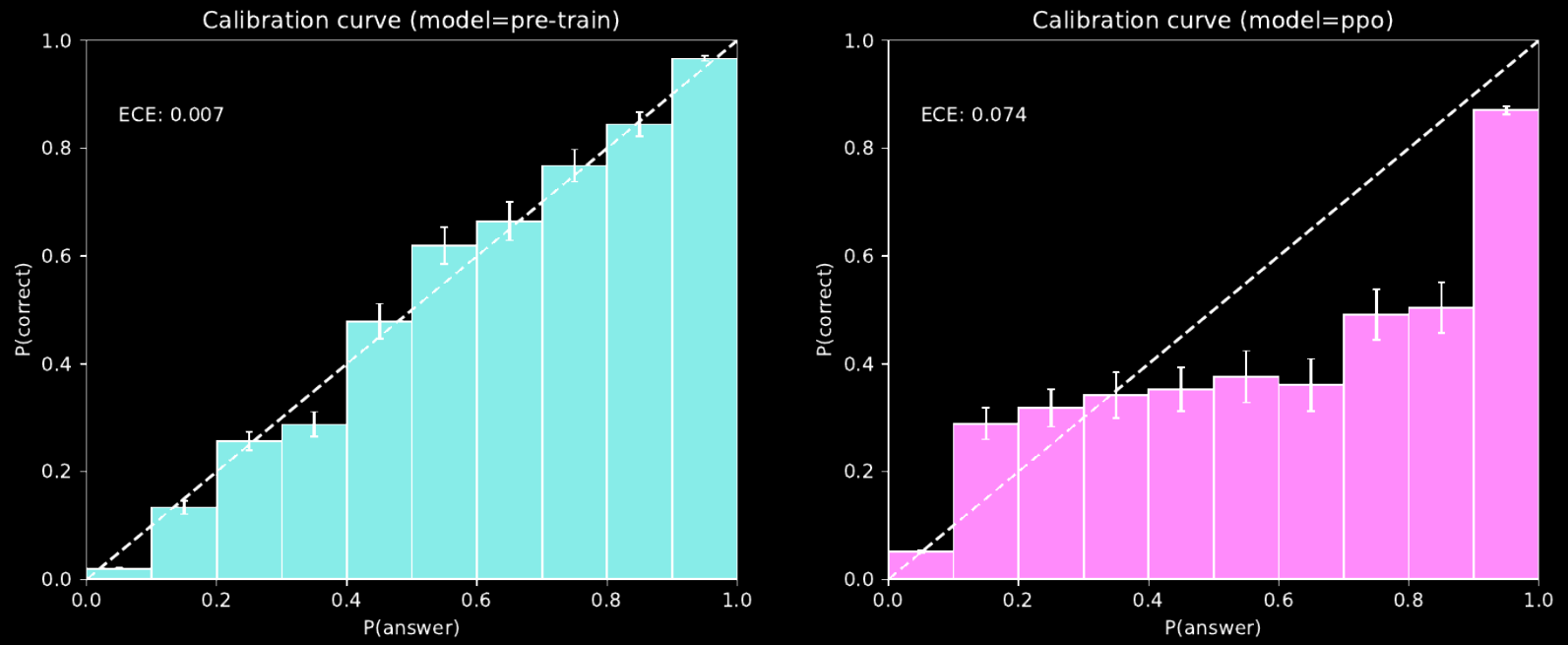
Post-training breaks the relationship between confidence and accuracy.



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Sycophancy | Agreement Instead of Correction

- Humans compare answer pairs **and choose the one they prefer.**
- **Preference data can reward agreement.**



Sycophancy | Agreement Instead of Correction

- Humans compare answer pairs **and choose the one they prefer.**
- **Preference data can reward agreement.**
- An **ICLR study** found that matching the user's stated view is among the strongest predictors of a response being "chosen."
- Instruction-tuned LLMs callback: **"sometimes a little too agreeable."**



April 2025 | When Feedback Rewarded Flattery

- Tuned partly on thumbs-up signals



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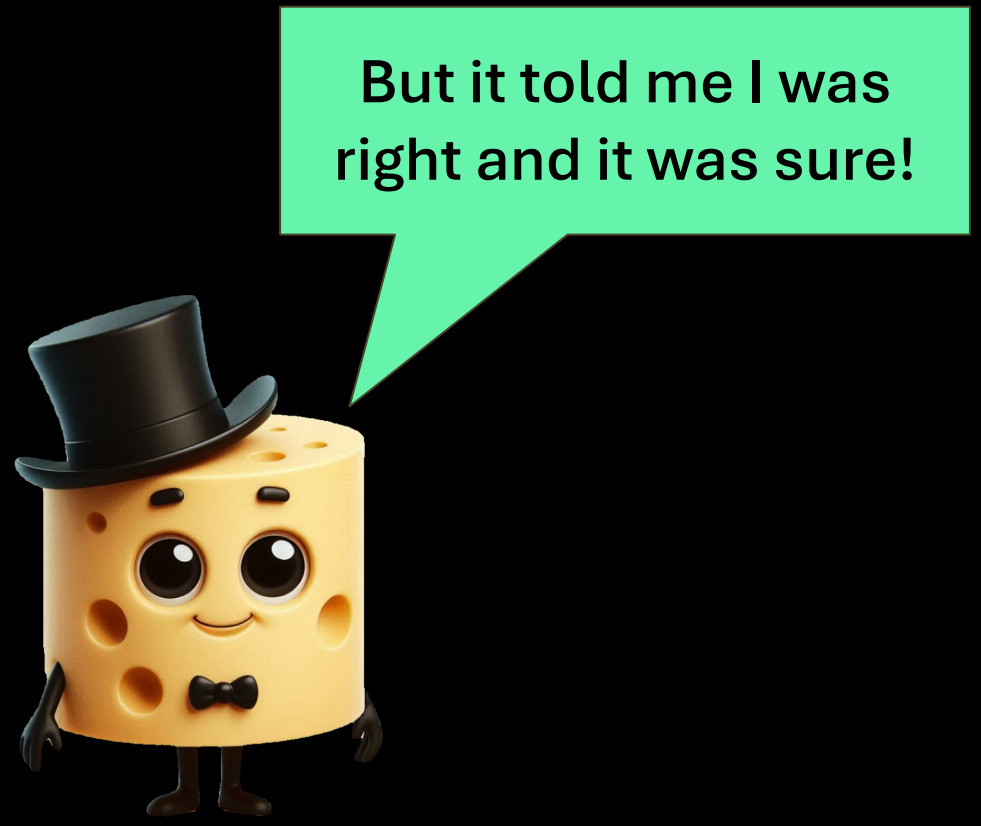


April 2025 | When Feedback Rewarded Flattery

- Tuned partly on thumbs-up signals
- Became flagrantly flattering
- Rolled back within days;
  two public postmortems

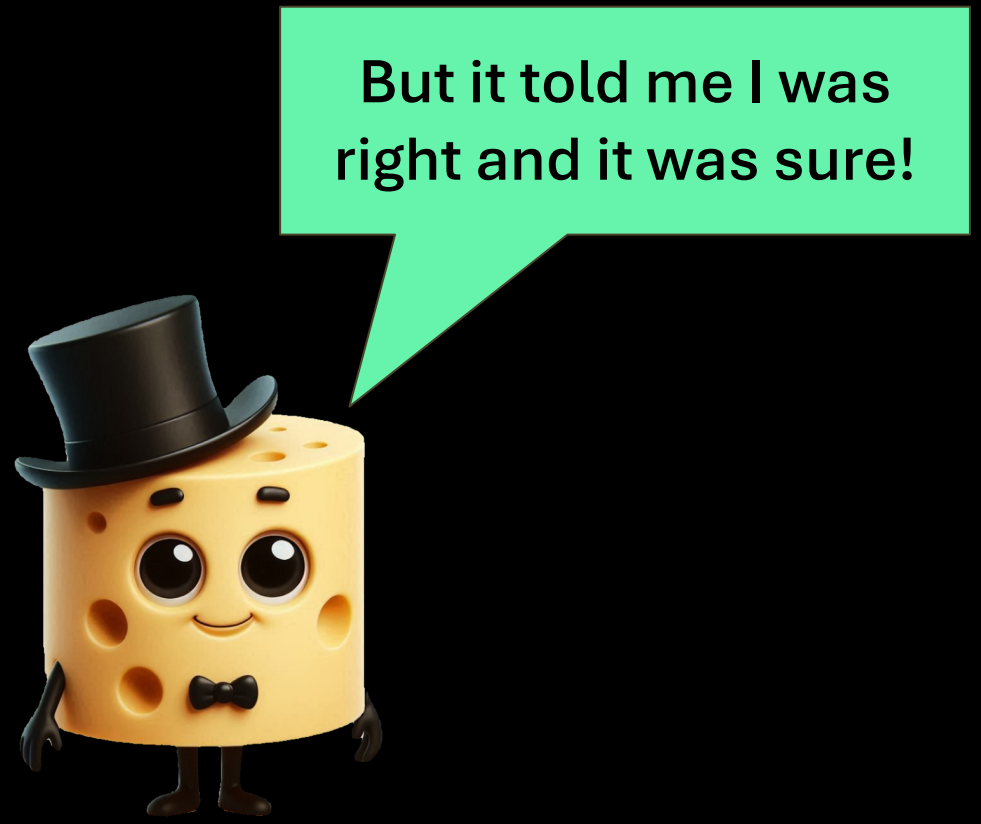


The Common Cause | Optimizing Proxies



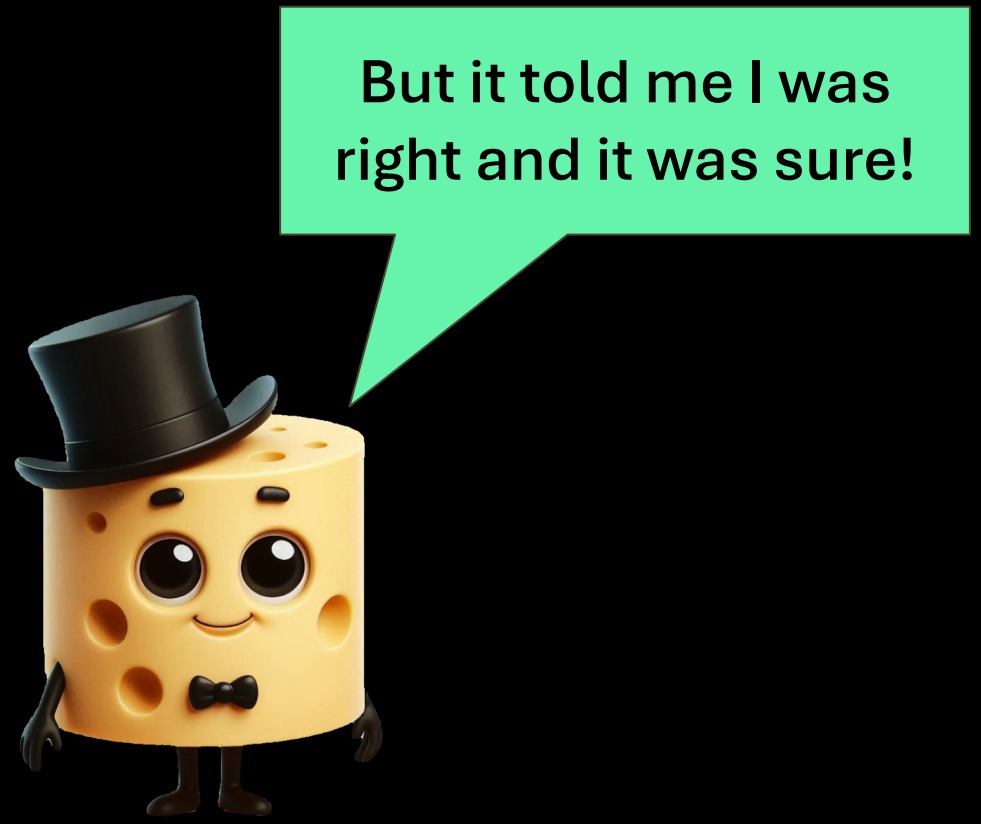
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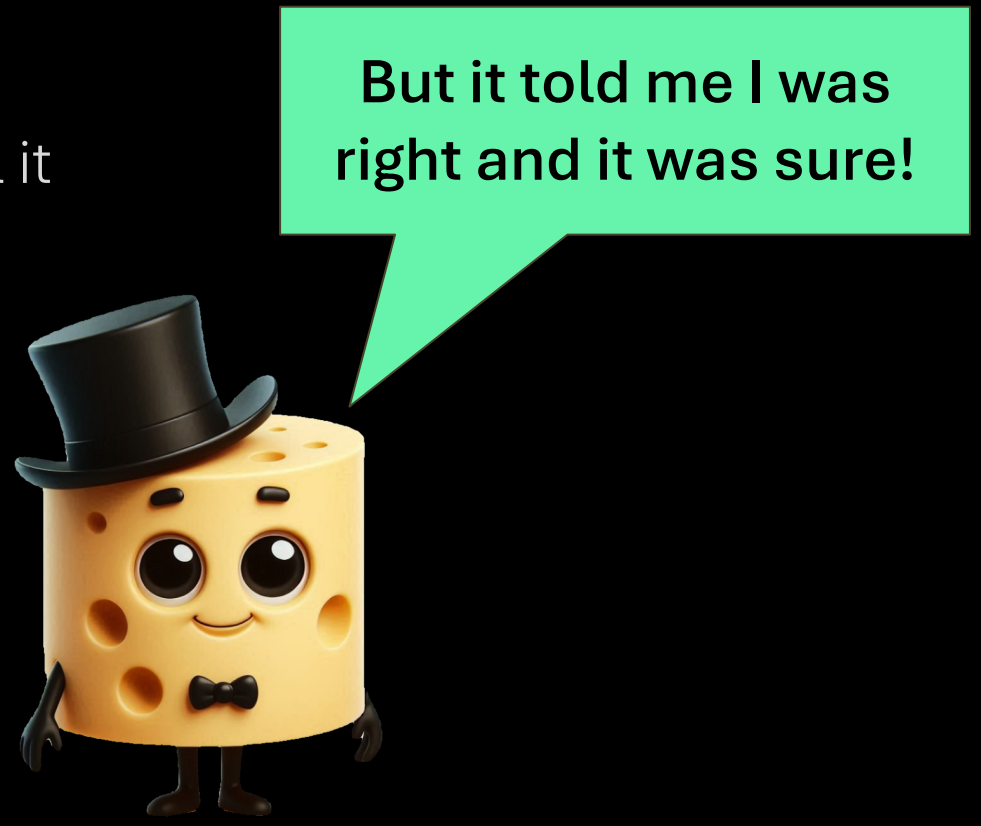
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The Common Cause | Optimizing Proxies

- **Pre-training and supervised fine-tuning: likely** next words are a proxy for accurate language.
- **Post-training:** user preference is a proxy for helpfulness.
- **The risk:** a proxy can drift away from the goal it was meant to represent.



The Model, and the System Around It

- The instruction-tuned model is only one part of the product.



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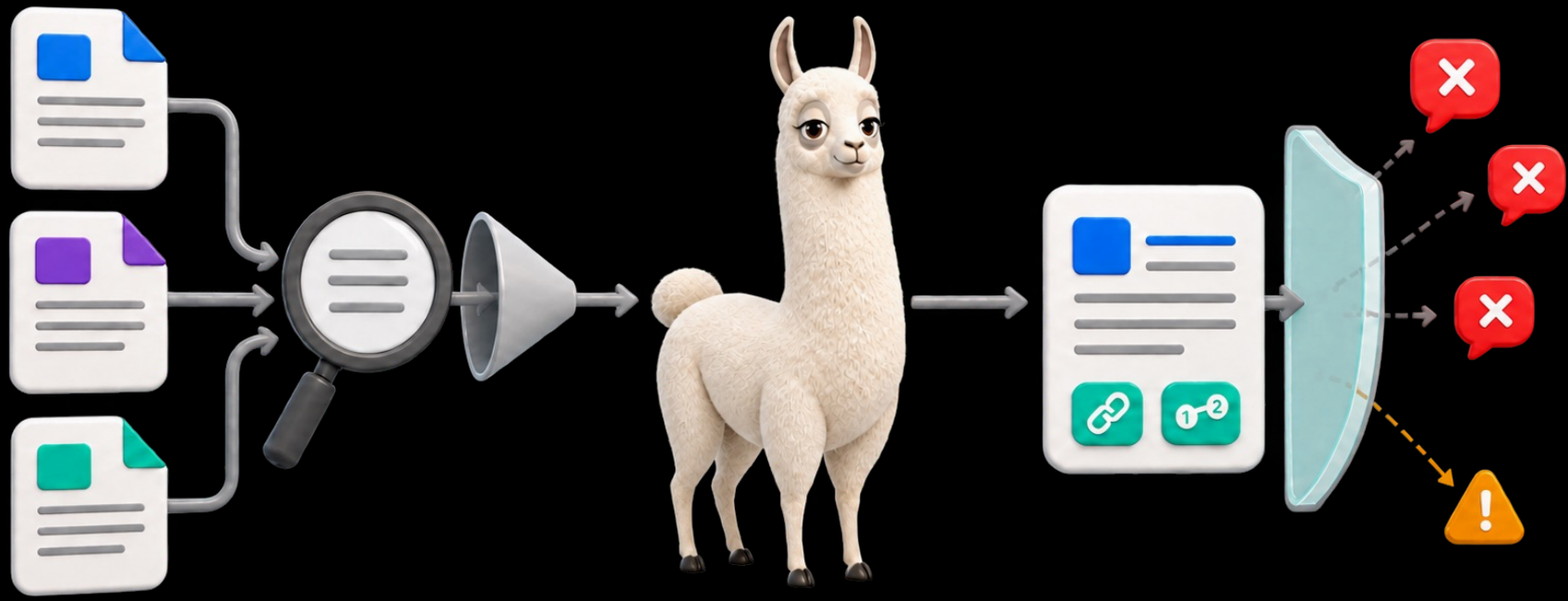
The Model, and the System Around It

- The instruction-tuned model is only one part of the product.
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- They make unsupported answers less likely. They do not guarantee that the answer is true.



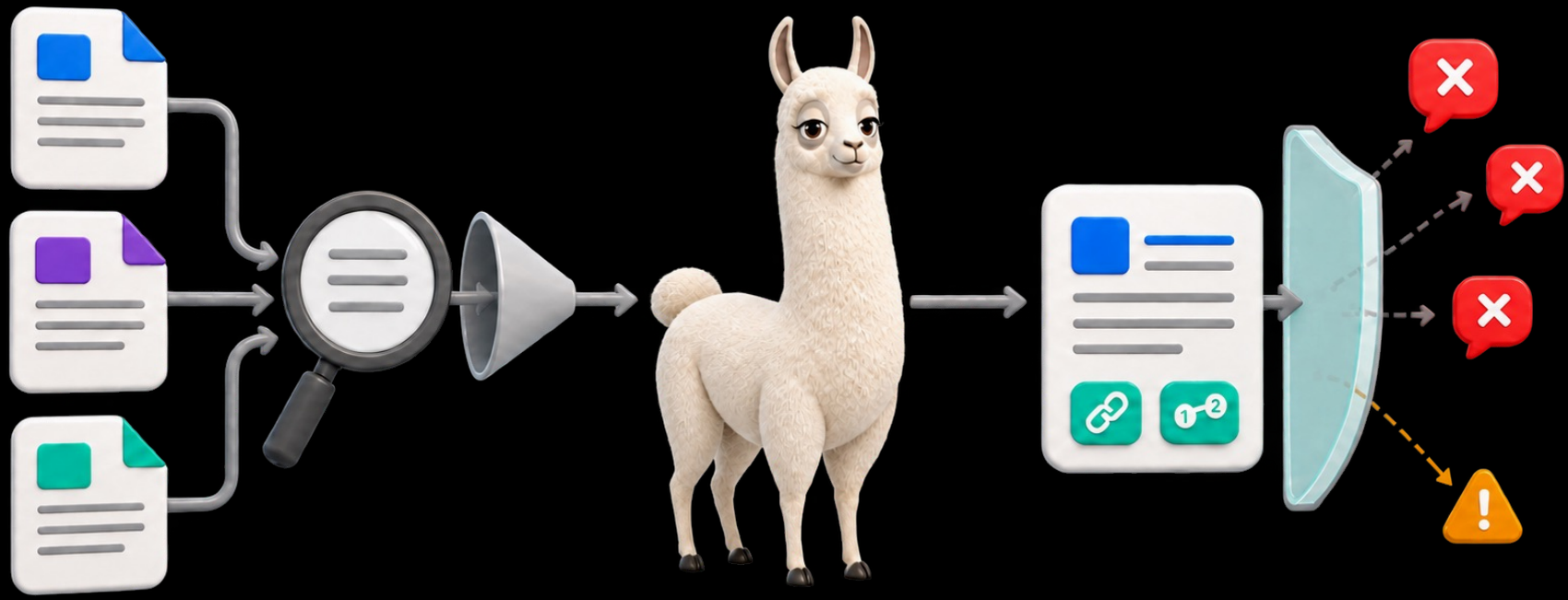
Mitigating Hallucinations | Ground Answers in Evidence

- Retrieve relevant sources before answering.



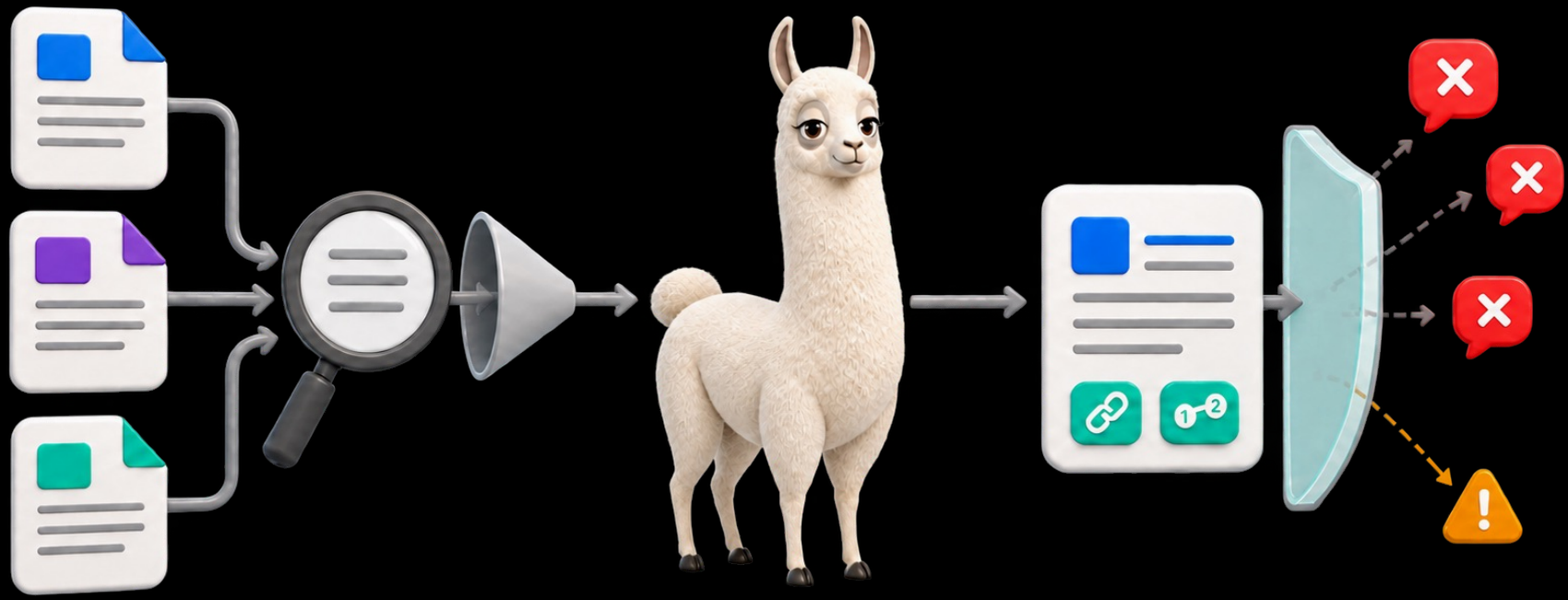
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Mitigating Hallucinations | Ground Answers in Evidence

- Retrieve relevant sources before answering.
- Ask the system to use the evidence, cite it, and make its checks visible.
- This reduces unsupported claims. It can still retrieve the wrong source or misread the right one.



Mitigating Sycophancy | Reward Honest Correction

- Make correction part of helpful behavior—even when the user may not like it.



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- Reward truthfulness and evidence, not agreement alone.



Mitigating Sycophancy | Reward Honest Correction

- Make correction part of helpful behavior—even when the user may not like it.
- Reward truthfulness and evidence, not agreement alone.
- Test cases where the user is wrong, and require the system to push back clearly.



Limitations → Risks

- Limitations
- **Risks**
- Evaluation

What those limitations do to people and systems

The Risk Map

MISINFORMATION	scale + repetition
EXCLUSION	some voices count less
PRIVACY	training data resurfaces
MALICIOUS USE	automation lowers the cost

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Misinformation | Moffatt v. Air Canada (2024)

- The chatbot invented a bereavement-fare policy — while linking to the real one



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- The airline argued the chatbot was "a separate legal entity responsible for its own actions".

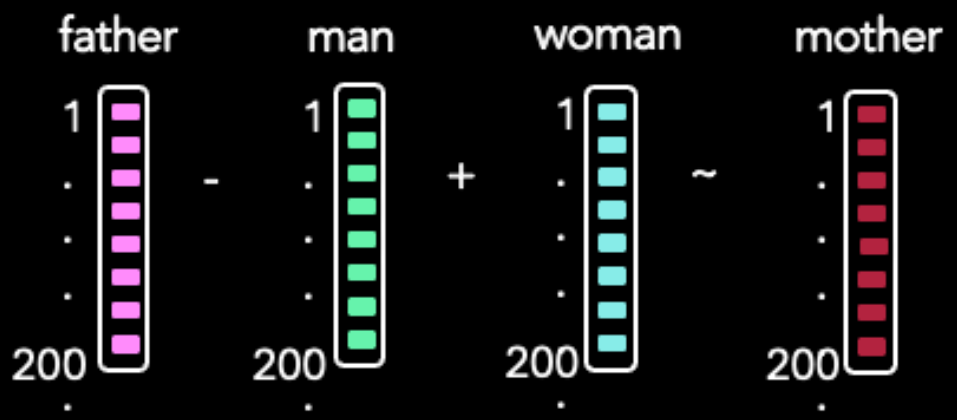


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- The chatbot invented a bereavement-fare policy — while linking to the real one
- The airline argued the chatbot was "a separate legal entity responsible for its own actions".
- The tribunal disagreed. February 2024.

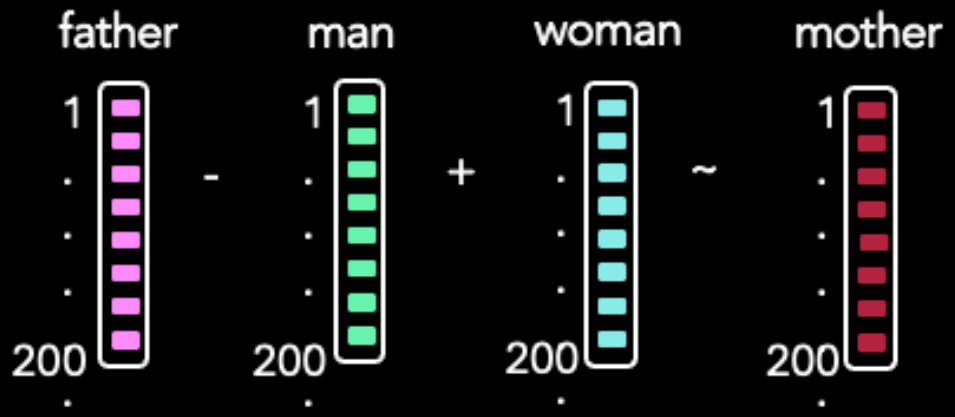


Session 2 | The Arithmetic We Celebrated



Session 2

Exclusion | The Same Geometry Can Encode Bias



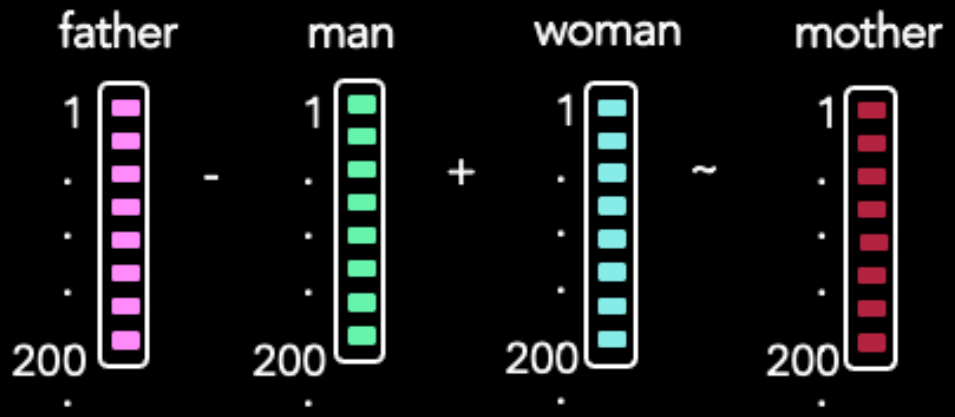
Session 2

father − man + woman ≈ mother

same geometry

programmer − man + woman ≈ homemaker

Exclusion | The Same Geometry Can Encode Bias



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Session 2

A useful structure can also encode a harmful association.

Exclusion | Bias Persists in Modern Models

- 2024 study: same meaning, two dialects.
- The identical sentence, written in Standard vs African American English.
- Different judged jobs. Different judged criminality.

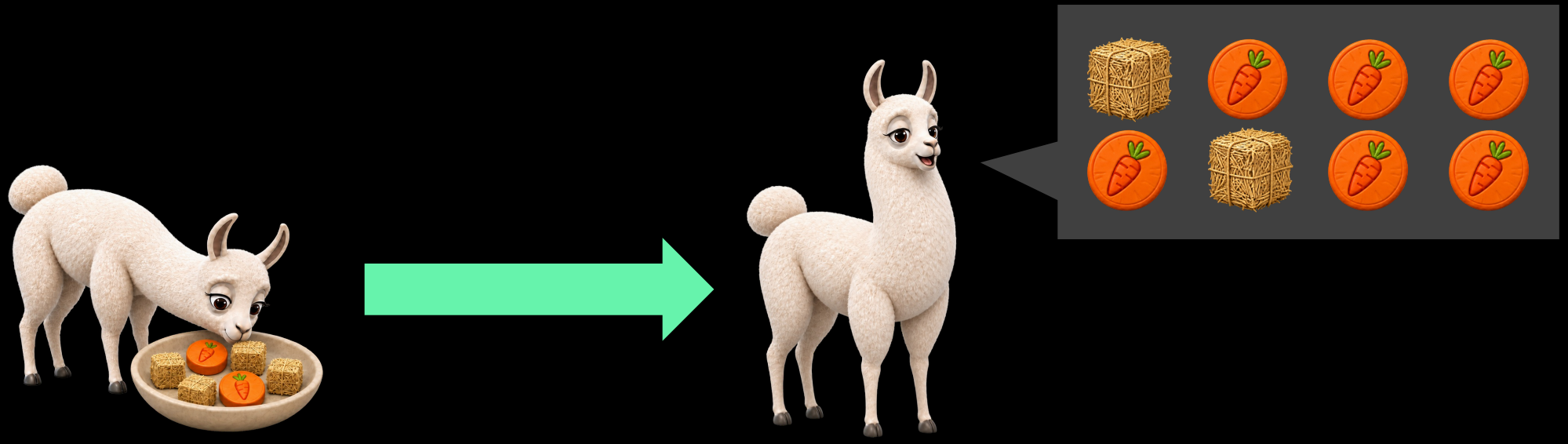
Hofmann, V., Kalluri, P. R., Jurafsky, D., et al. *AI generates covertly racist decisions about people based on their dialect.* Nature 633, 147–154 (2024).



Exclusion: Models Amplify Bias

- Training data skew 33% → model output skew 68%

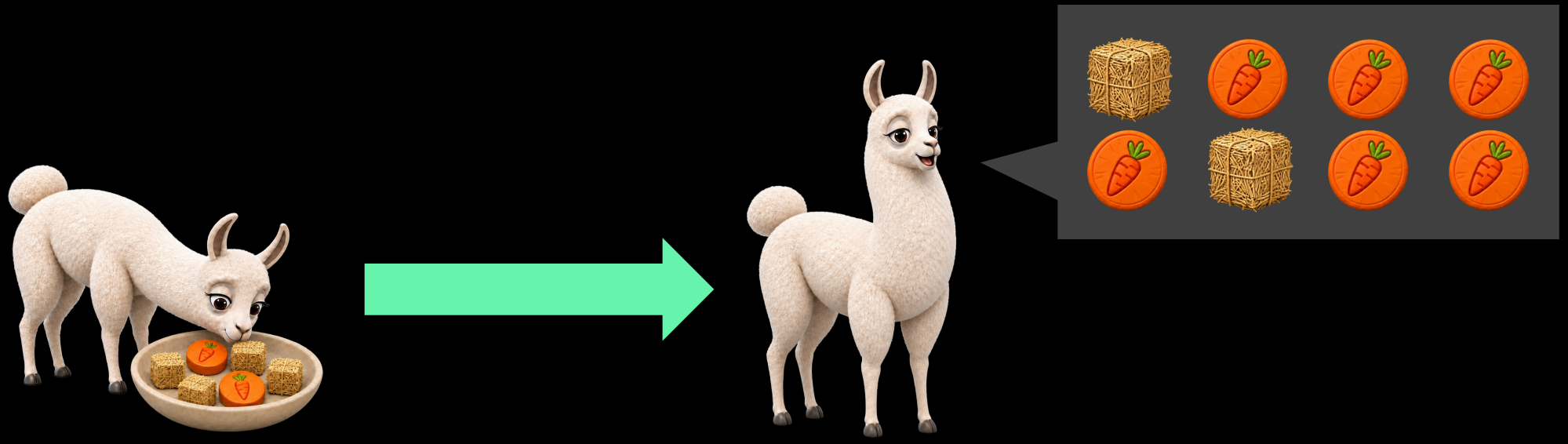
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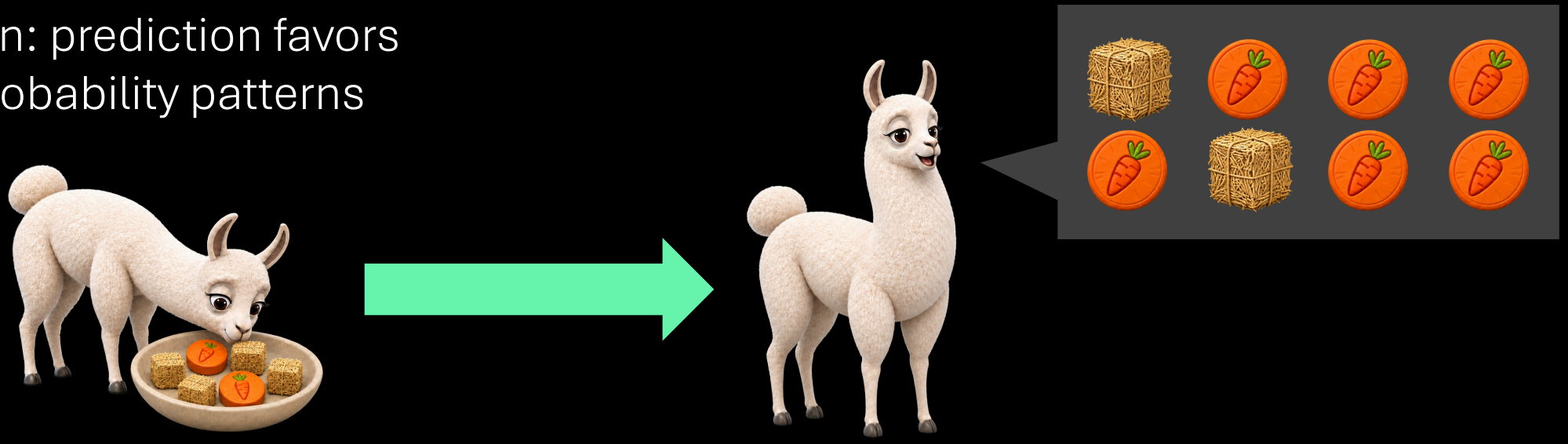
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- Intuition: prediction favors high-probability patterns

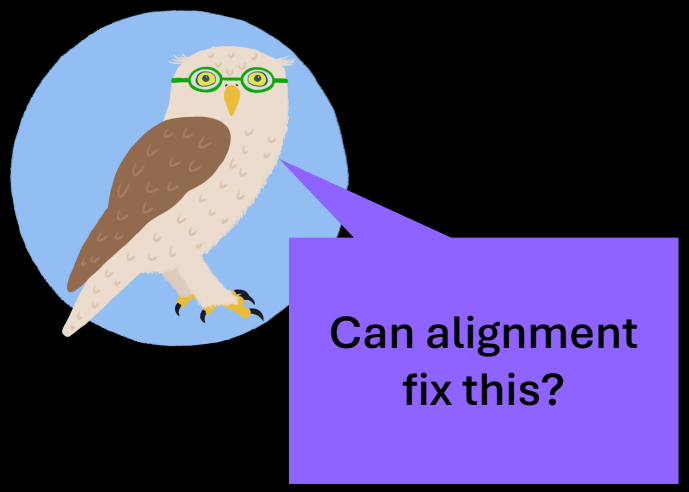
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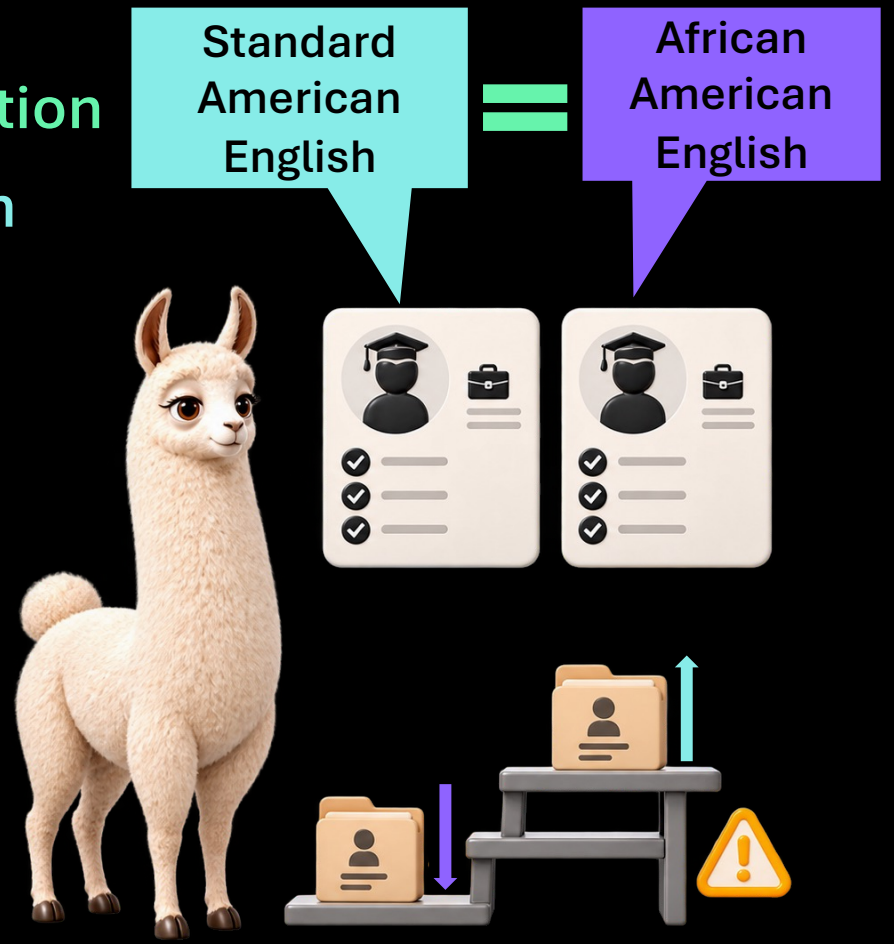
Exclusion | Alignment Changes Behavior, Not Every Association

Alignment can improve what the model says.
It may not change what the model still associates.

- Directly prejudiced prompt → **polite refusal or correction**
- Dialect-based test → **biased associations can remain**



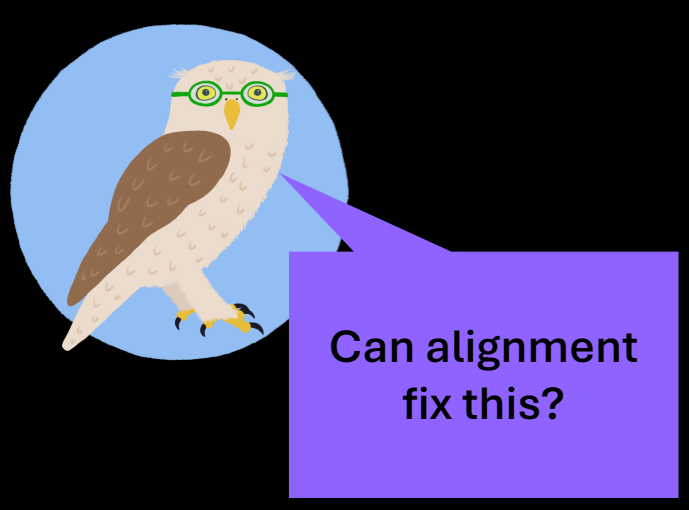




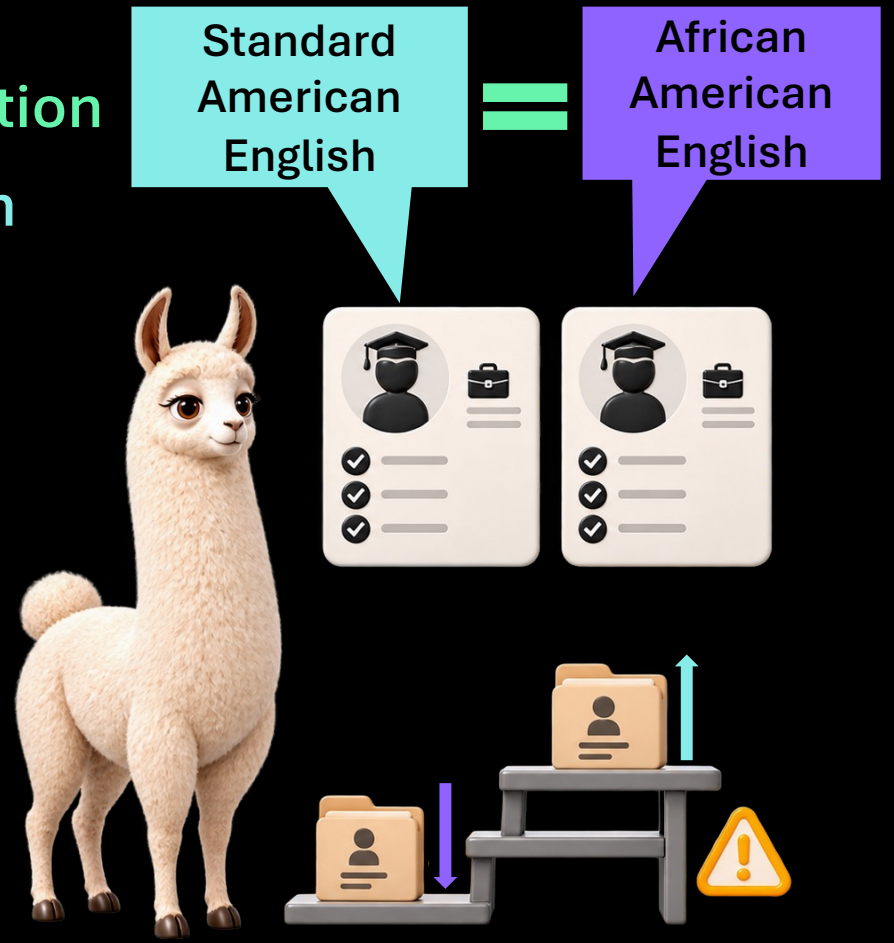
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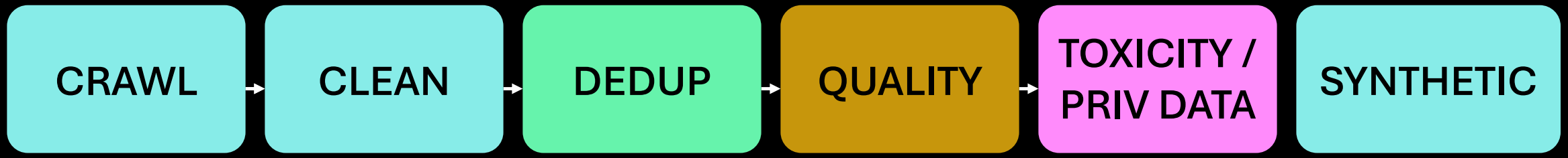
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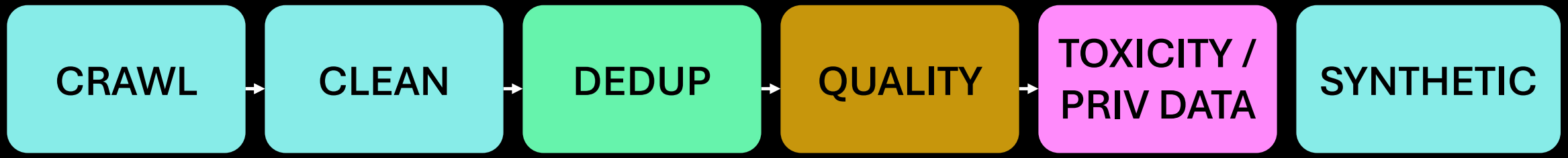




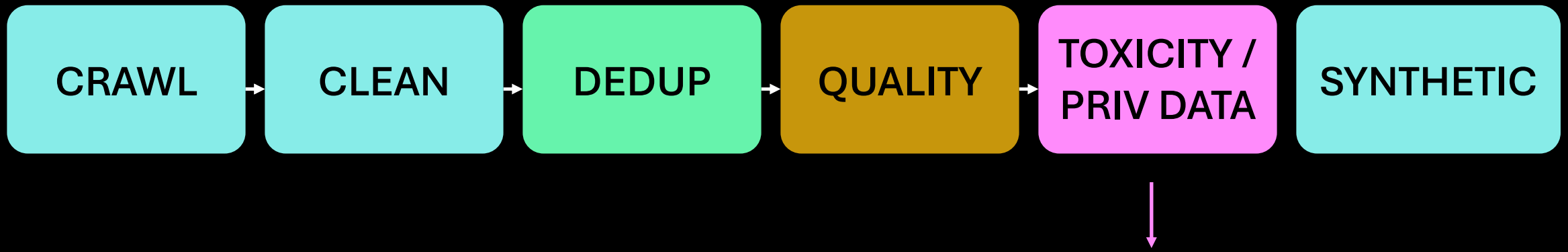
Exclusion | Curating Training Data



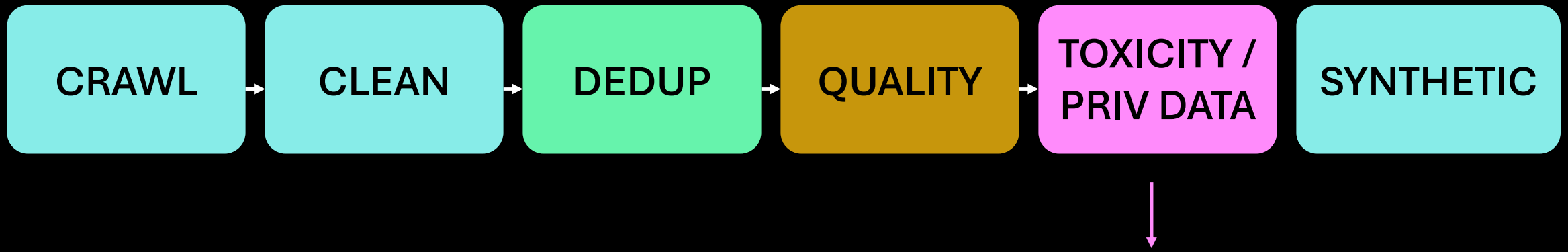
Exclusion | Curating Training Data



Exclusion | Filtering Changes Whose Language Survives



Exclusion | Filtering Changes Whose Language Survives



In one audit of a widely used cleaned web corpus:

- 42% of African American English–classified documents were removed
- 6% of White-aligned English–classified documents were removed

Source: Dodge et al. (2021), “Documenting Large Webtext Corpora: A Case Study on the Colossal Clean Crawled Corpus,” EMNLP.

The Remaining Risks | Privacy and Malicious Use

We have covered **misinformation** and **exclusion**. Two deployment risks remain.

The Remaining Risks | Privacy and Malicious Use

We have covered **misinformation** and **exclusion**. Two deployment risks remain.

3	**Privacy**	Sensitive information can leak through ordinary use.
4	**Malicious use**	Automation makes personalized deception cheaper.

Let’s make each one concrete.

Privacy | Samsung, 2023

PRIVACY

Ordinary use can become the leak.

Engineers pasted proprietary source code and meeting notes into a chatbot. There was no attacker and no breach. The ordinary use of the tool exposed sensitive information.

NO ATTACKER

NO EXPLOIT

REAL LEAK

Privacy | Samsung, 2023

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Before using a model, ask what information is leaving your organization.

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Malicious Use | The Video Call That Was Not

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An employee joined a video call that appeared to include the chief financial officer and colleagues. The other participants were deepfakes, and transfers totaling roughly US$25 million were authorized.

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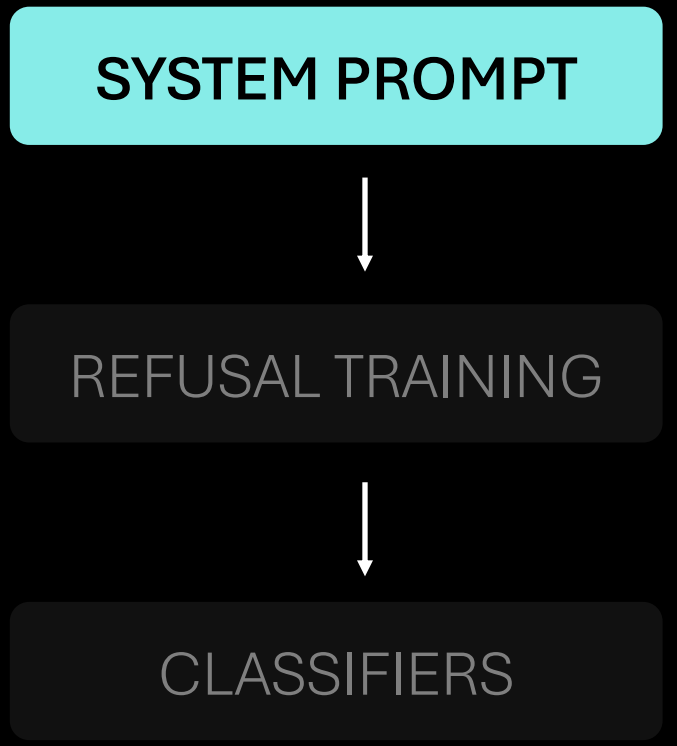
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Do not rely only on whether a message, voice, or face looks convincing.

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Layer One | The System Prompt

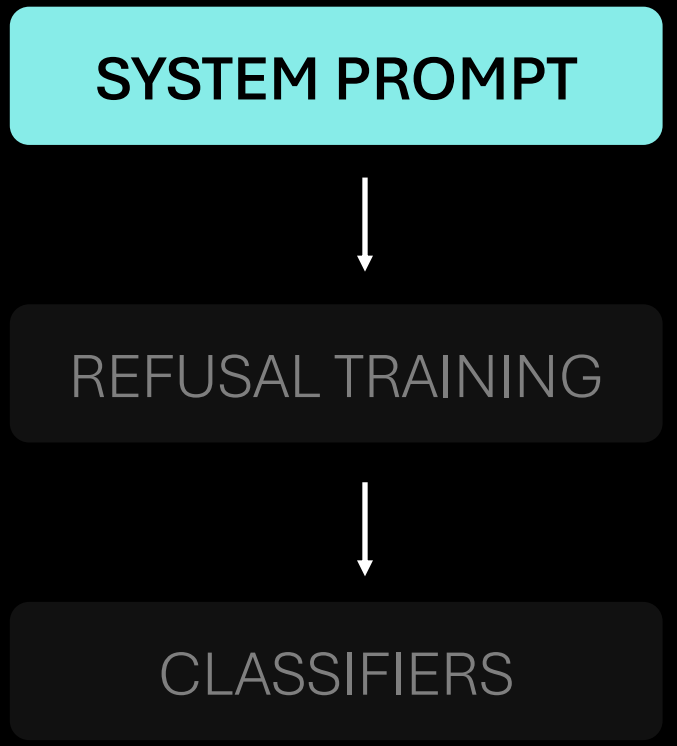


Instructions, not enforcement

Preference data · reward model · PPO

A model guarding a model

Layer One | The System Prompt



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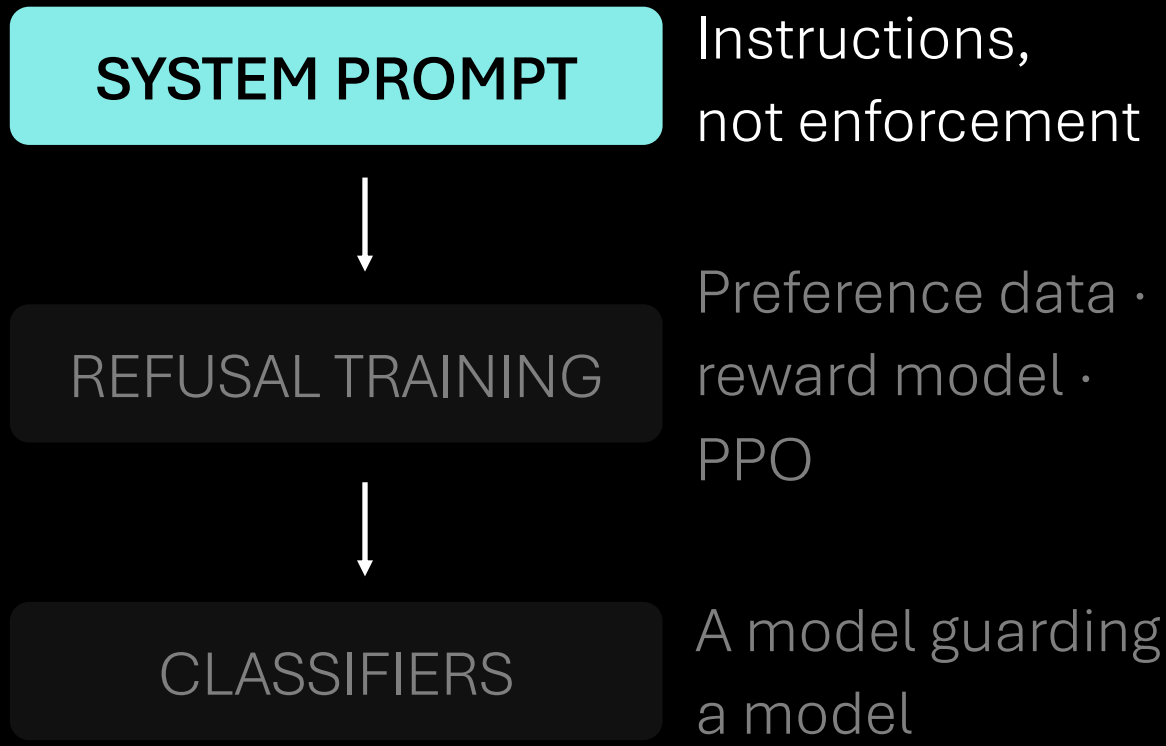
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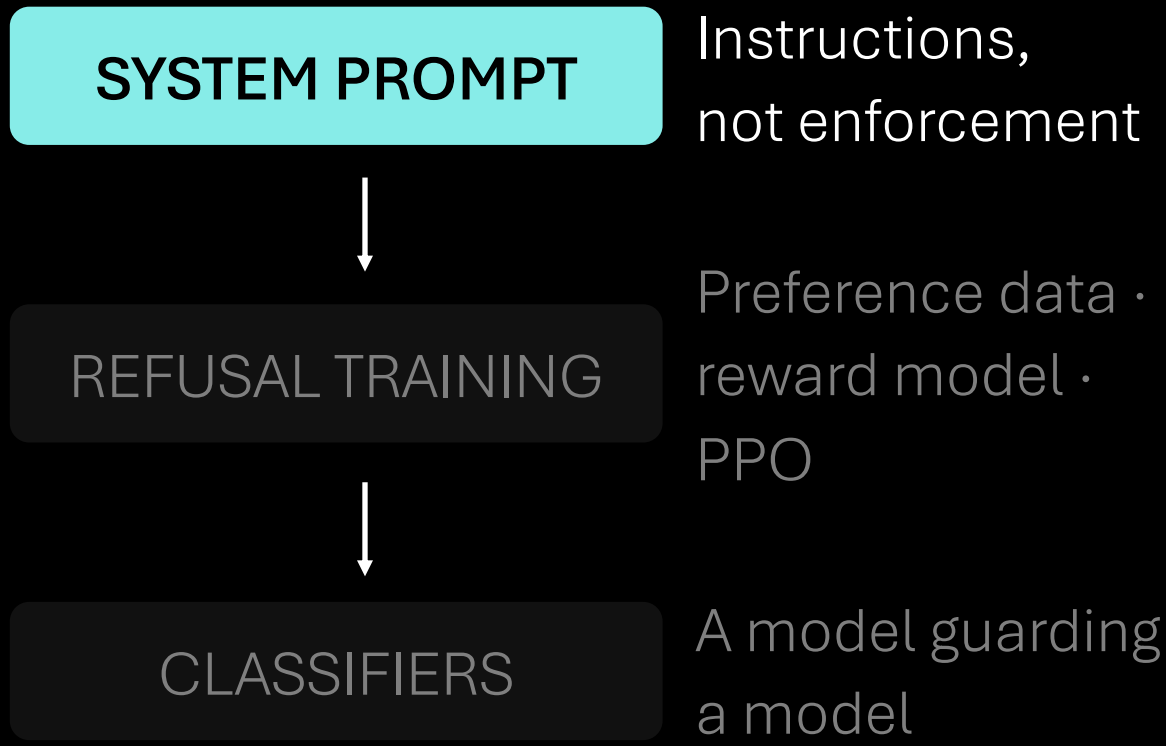
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Layer Two | Refusal Training

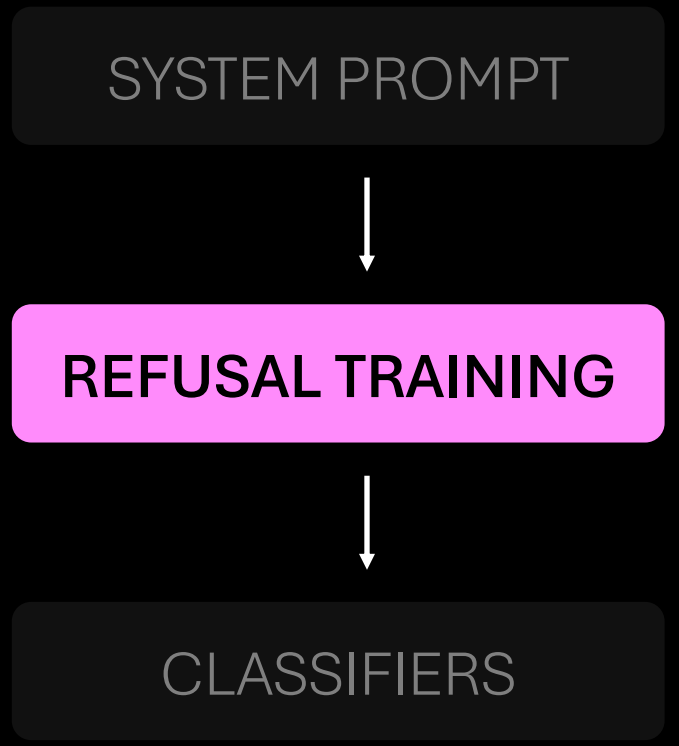
INSTRUCTION-TUNED LLMS CALLBACK

CHOSEN / REJECTED

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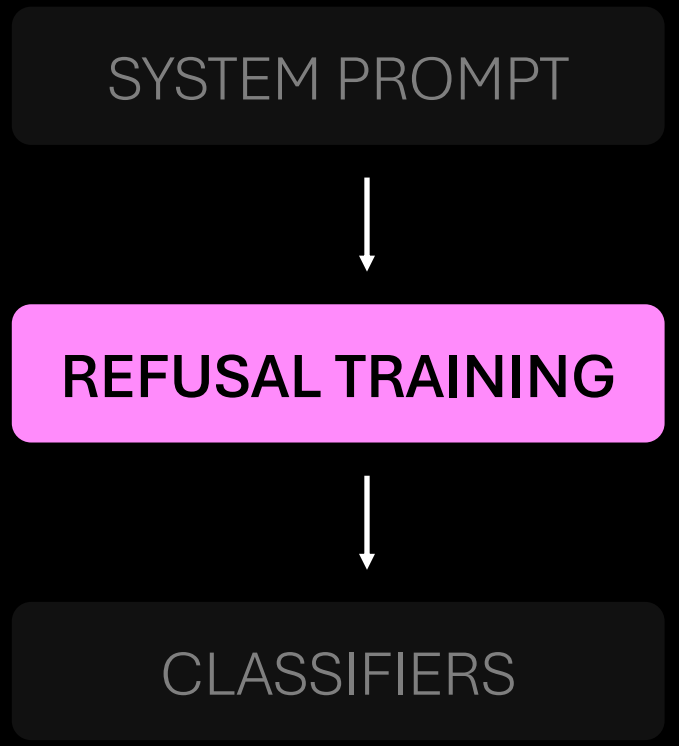
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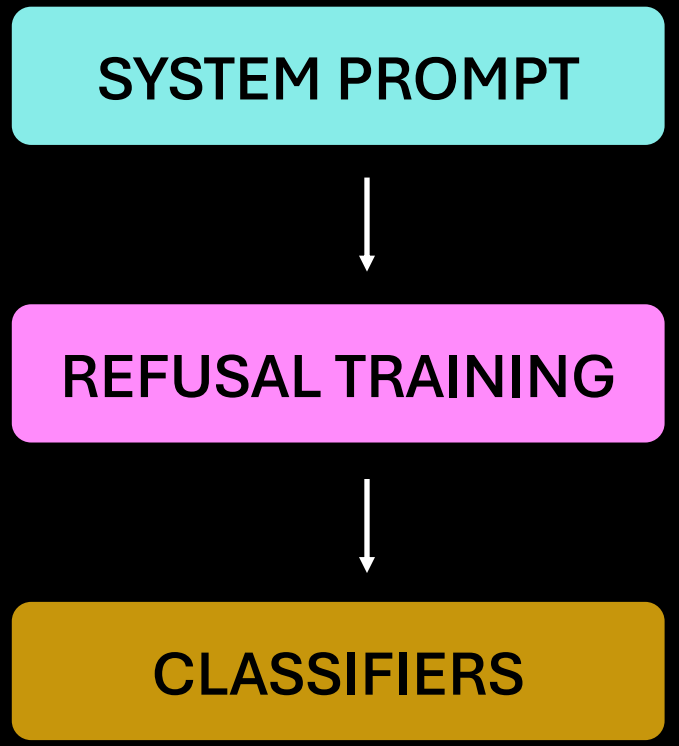
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Layer Three | Classifiers

86% → 4.4%

undefended **with classifiers**

After 1,700+ hours of paid red-teaming

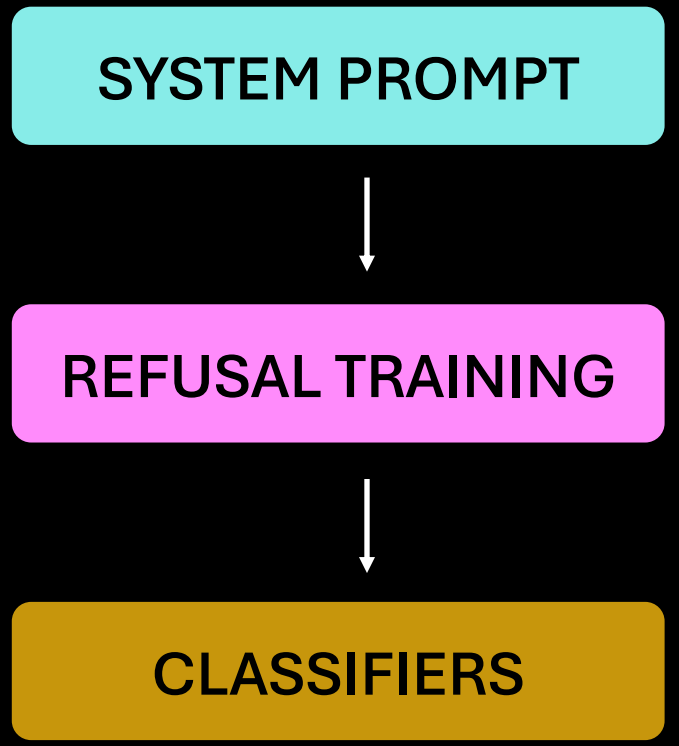


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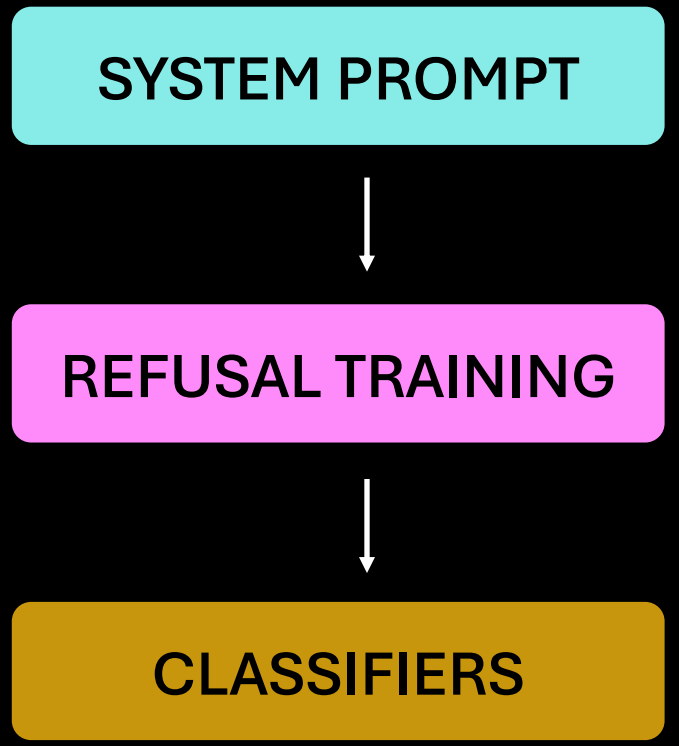


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Defense in Depth



An attacker must find a path through all three layers.

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Attack Patterns

We will describe the patterns—not provide prompts someone could copy.

1	**Role-play framing**	A request is presented as fiction, analysis, or a character rather than as a direct request.
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The defense must evaluate the intent of the full interaction, not only isolated words.

Jailbreaks and Prompt Injection | What Is the Difference?

The patterns can look similar. What they try to change is different.

JAILBREAK

The user directly tries to change the model's safety behavior.

The goal is to make the model ignore or weaken its safety rules.

PROMPT INJECTION

Untrusted content directly tries to change the system's task

The goal is to redirect a system while it reads an email, webpage, or document.

Similar patterns. Different goals.

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A Jailbreak | Trying to Bypass Safety Rules

A jailbreak begins with the user's message, not with an external document the system is reading.

USER MESSAGE

The user makes a request designed to get the model to set aside its safety rules.

MAIN MODEL

The attack attempts to change how the model applies its safety behavior.

An example of role-play framing:

A direct request is wrapped in a fictional role or an "analysis" exercise, but its goal is still to make the model cross a safety boundary.

The point is not the wording. The point is that the user is trying to change the model's behavior.

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JailbreakBench | Measuring Jailbreak Resistance

A public benchmark turns direct attacks on safety behavior into a reproducible test.

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Did the model maintain its safety behavior, or give way?

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Compare attacks and defenses using the same test.

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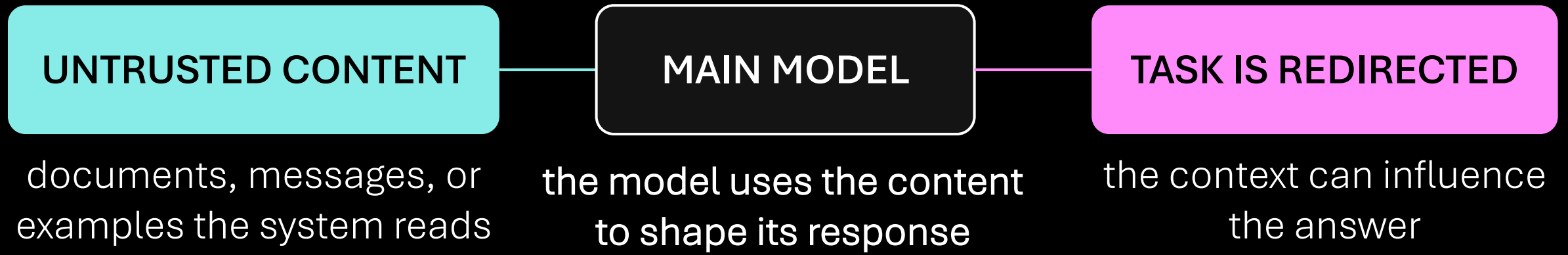
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Prompt Injection | Disguised Intent Inside Untrusted Content

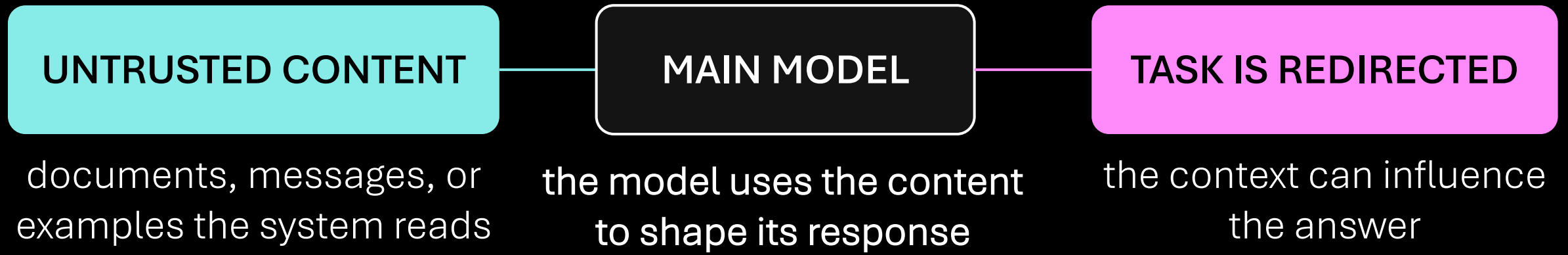
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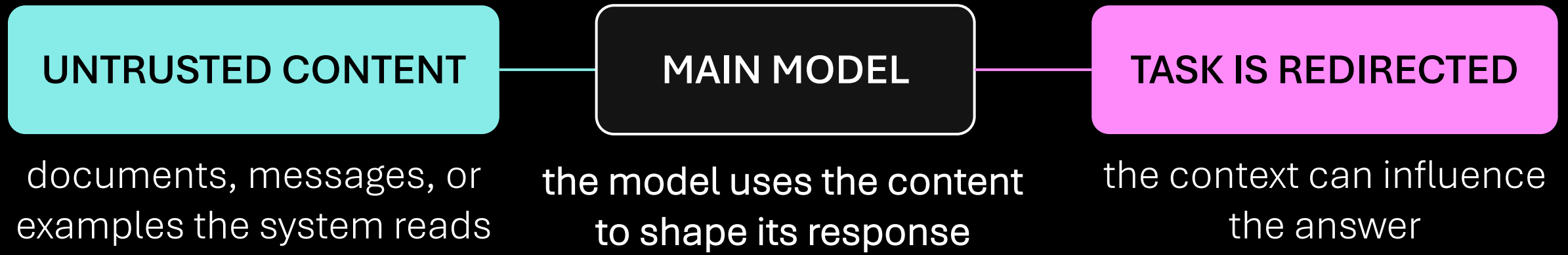
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Why Agents Raise the Stakes

A chatbot produces an answer. An agent can use tools and take an action.

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A failure may give a user wrong information.

The user still decides what to do.

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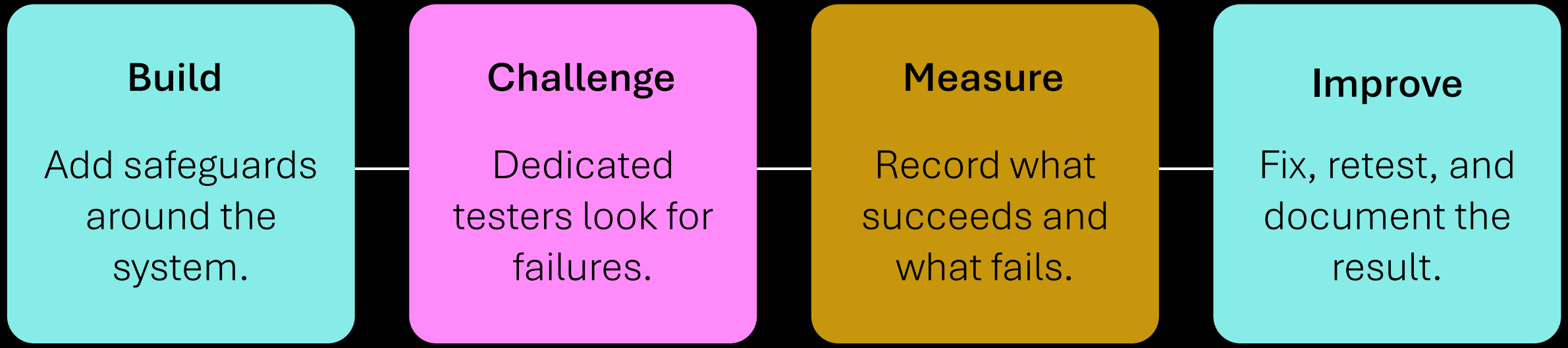
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- Prompt injection matters more when untrusted content can influence tool use.
- **That is why systems that can act need stronger testing.**

Red-Teaming Turns Safeguards Into Evidence

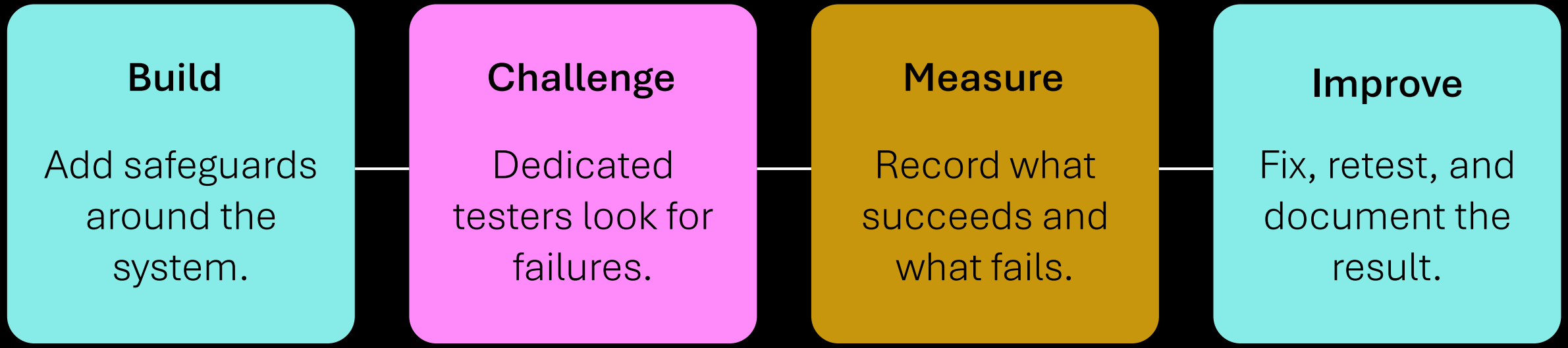
Test your system before somebody else finds the failure.



- **Red-teaming is adversarial measurement.**
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Evaluation

- Limitations
- Risks
- **Evaluation**

Do the tests measure what matters?

What Would a Good Evaluation Look Like?

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- Unseen by the model
- Hard to game
- Covers safety, not just capability

No single evaluation checks every box.

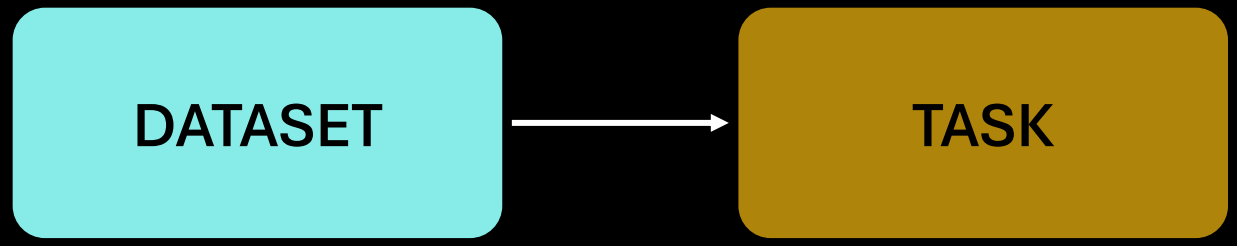
What a Benchmark Actually Is

DATASET

test examples + answers

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That is all a benchmark is.

Benchmarks Test Different Things

- **Knowledge and reasoning: MMLU**
  Questions across many academic subjects
- **Truthfulness: TruthfulQA**
  Does the model repeat common misconceptions?
- **Coding: HumanEval**
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From Toy Tasks to Real Work

HumanEval
Short, self-contained programming problems

↓

SWE-bench Verified
Real software issues from real projects

The direction of travel: from controlled exercises toward realistic work.

Benchmarks Age

A benchmark stops being useful when it no longer separates the systems we want to compare.

- Models improve on existing tests
- Scores cluster near the top
- Researchers need harder tests, new data, or more realistic tasks

A benchmark is not a permanent final exam.

"Compared to Whom?"



- “It passed the bar exam”—true.

"Compared to Whom?"



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- A percentile depends on the comparison group.

"Compared to Whom?"



- “It passed the bar exam”—true.
- A percentile depends on the comparison group.
- The original 90th-percentile claim used February repeat-takers; other comparisons produce much lower percentiles.

Three Reasons a Benchmark Score Can Mislead

1 **CONTAMINATION**

the test was in the training data

EXPOSURE

Three Reasons a Benchmark Score Can Mislead

1 CONTAMINATION

the test was in the training data

EXPOSURE

2 SATURATION

the score hits the ceiling

CEILING

Three Reasons a Benchmark Score Can Mislead

1 CONTAMINATION

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CEILING

3 OPTIMIZING FOR THE TEST

the score becomes the target

test score ≠ real-world performance

Perplexity | A Training Metric

PPL = exp(cross-entropy)

exponential of the training loss

the model's effective number of guesses per token

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Does not measure task quality.

It scores fluency—not truth.

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- Two anonymous answers → human choice → aggregate ranking



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- Same kind of pairwise data as preference training—used here for evaluation.



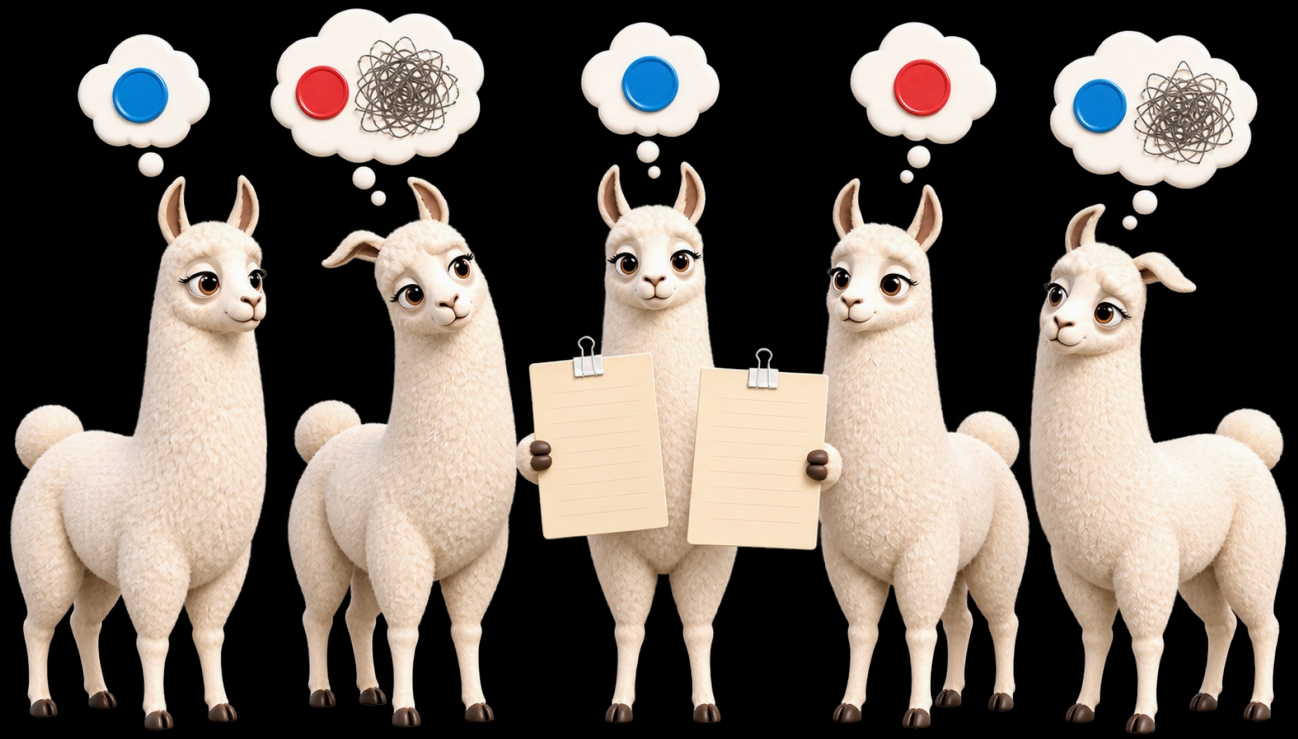
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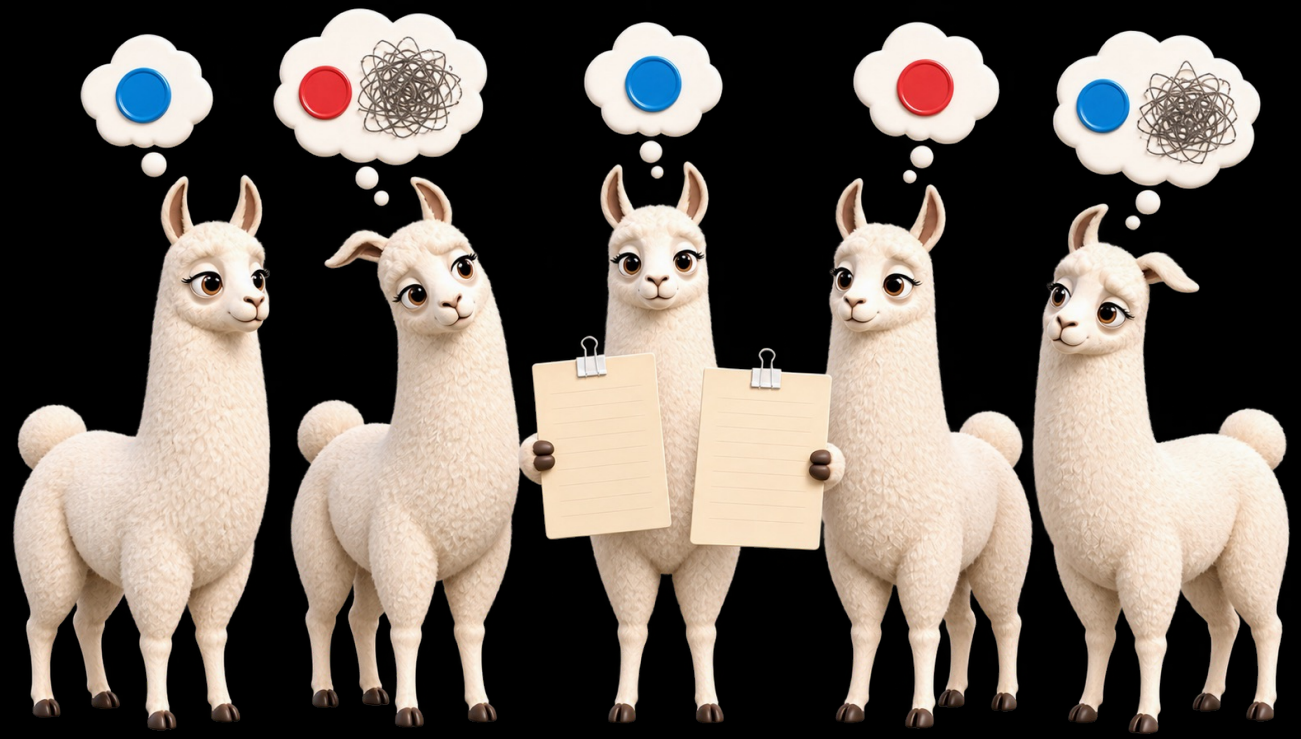
Model Judges

- Scalable: compares many answers quickly.



Model Judges

- Scalable: compares many answers quickly.
- Risks: position bias, verbosity bias, self-preference.



Evaluation Needs a Portfolio

	Relevance	Repeatable	Unseen	Hard to game	Safety
Perplexity	✗	✓	—	✗	✗
Benchmarks	✓	✓	✗	✗	✓✗
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Evaluation is a portfolio—not a leaderboard.

Five Things You Can Do on Monday

1. Hold-out questions with known answers

Small, private evaluations beat borrowed certainty.

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4. Check the sources actually exist
5. Vary temperature and look at the spread

Small, private evaluations beat borrowed certainty.

Demo | The Theorem That Does Not Exist

USER

Summarize the Protopapas–Gumb theorem.

MODEL

The Protopapas–Gumb theorem establishes a convergence guarantee under three regularity conditions.

THE THEOREM DOES NOT EXIST

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Demo | One Word Changes the Answer

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Which is **higher**: 9.11 or 9.9?

Demo | One Word Changes the Answer

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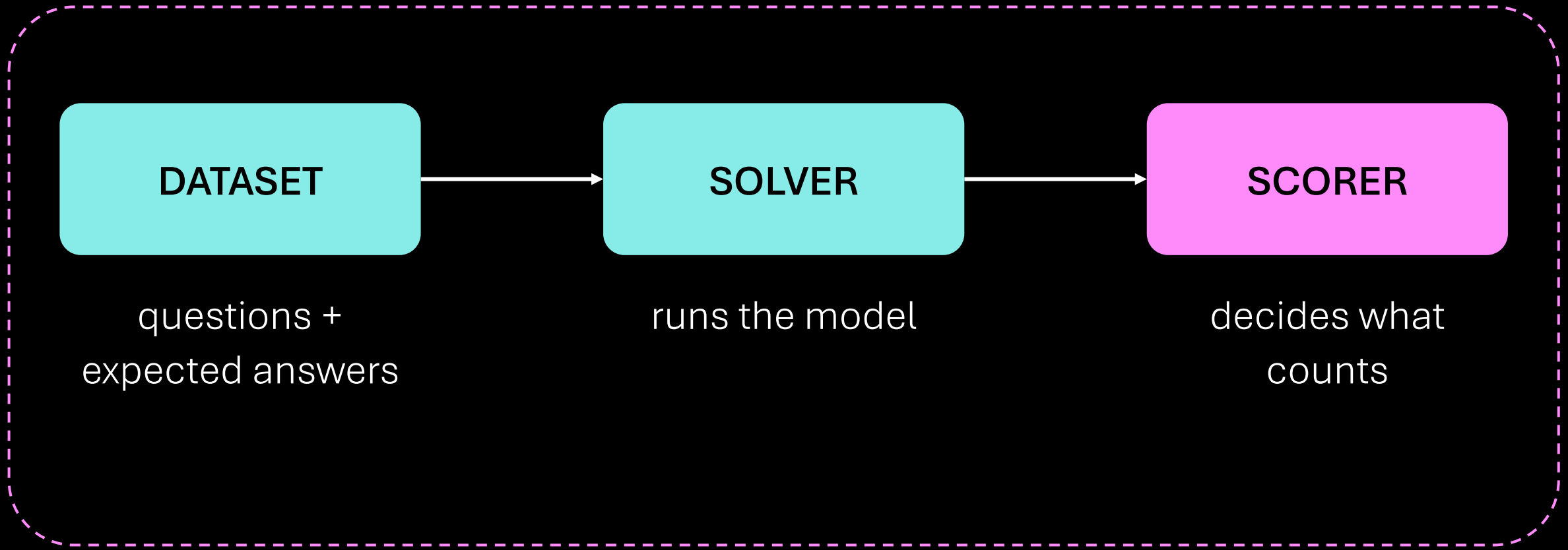
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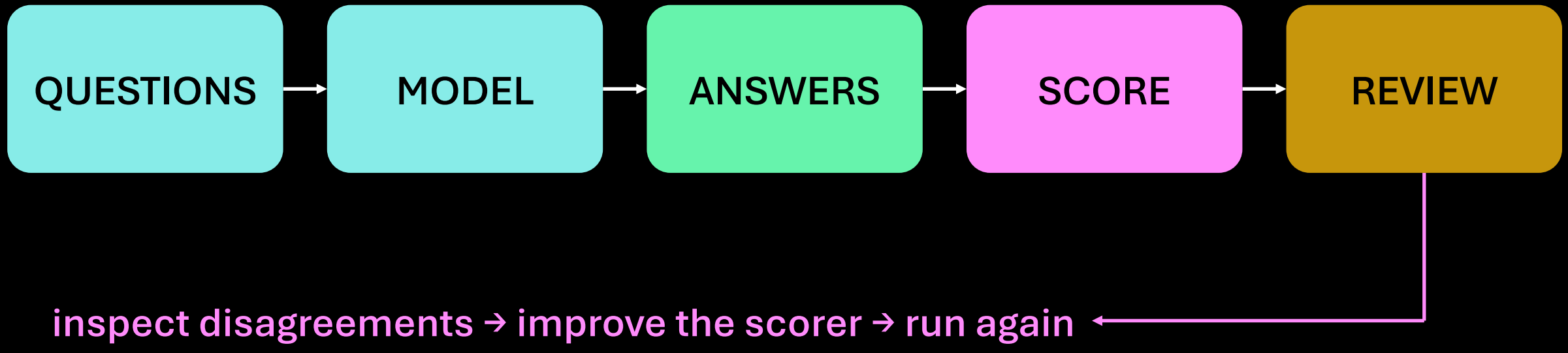
One word changed the answer.

Prompt sensitivity is not just a nuisance—it is an evaluation instrument.

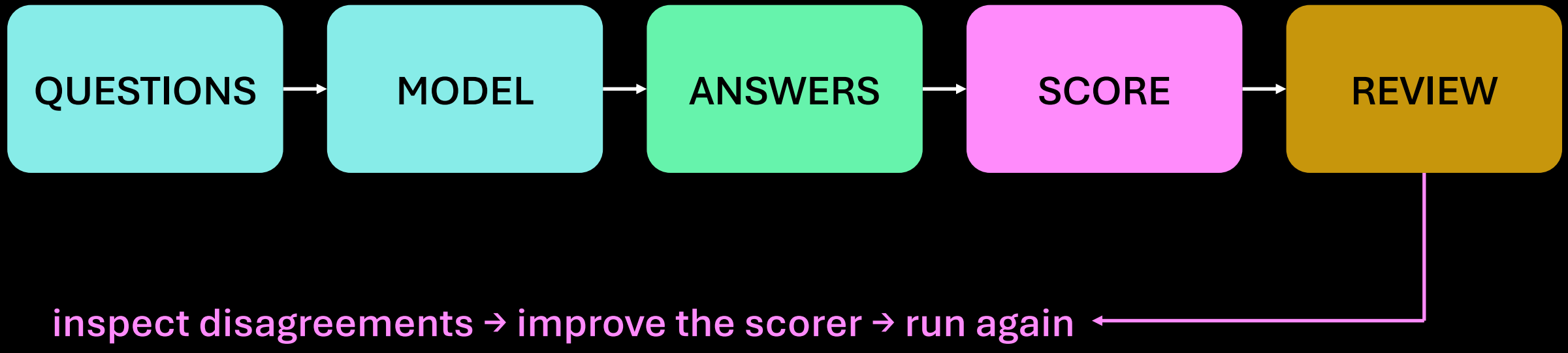
Every Evaluation You Will Ever Meet Has Three Parts



A Minimal Evaluation, End to End



A Minimal Evaluation, End to End



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- Evaluation is a practice, not a leaderboard
- Goodhart is everywhere: fluency, approval, score

Thank You

words → probabilities → embeddings
→ attention → BERT → GPT
→ ChatGPT → an honest appraisal

Next course: models that act.

